Block practice · Cells, DNA and cell division

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Compare light and electron microscopes and use their parts

Draw or label a human cell as seen under the microscope

Name the parts of a nucleotide and the DNA backbone

Define specialised cells, tissues and organs and identify human tissues

Match human cell structures to their functions

Prepare cheek-cell or blood slides for a light microscope

Compare DNA with RNA

Describe DNA's double helix, base pairing and replication

Describe the products and purposes of mitosis

Describe transcription and translation in human cells

Apply the triplet code to bases, codons and amino acids

Use image size, actual size and magnification

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Practice workbook · Compare light and electron microscopes and use their parts

Why magnification or an electron microscope helps, which organelles only an EM reveals, and functions of light microscope parts.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-COMPARE-LIGHT-ELECTRON-MICROSCOPE

1. Question 3 · part c · 2 marks

June 2023 Paper 2R

Worked-example retry · Attempt part c only.

3 (a) Describe what is meant by a specialised cell.

(2)

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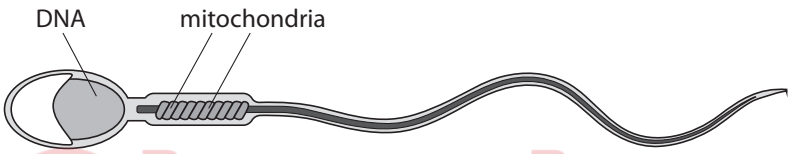
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE



(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

(2)

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

(1)

.....

.....

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1. Question 3 · part c · 2 marks

June 2023 Paper 2R

Worked-example retry · Attempt part c only.

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(iii) Explain why the sperm cell has many mitochondria.

(2)

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(iv) A sperm cell travels at a speed of 0.070 mm/s.

It travels a distance of 180 mm to fertilise an egg.

Use this equation to calculate the time it takes for the sperm to reach the egg.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Give your answer in minutes.

(3)

time taken = ... minutes

(c) An electron microscope can be used to view structures in specialised cells.

Explain why an electron microscope is more useful than a light microscope to view cell structure.

(2)

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(Total for Question 3 = 12 marks)

1. Question 3 · part c · 2 marks · Working and answer

Working and answer

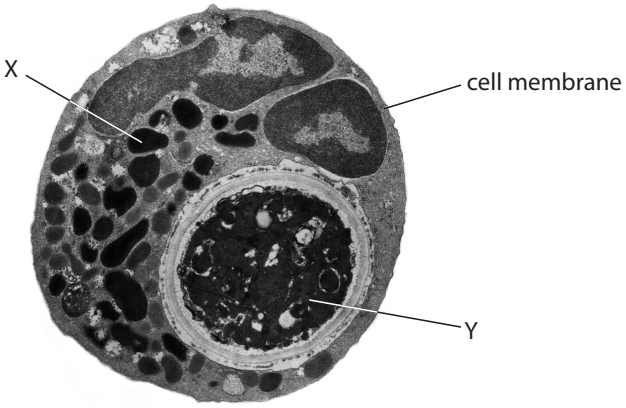
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2. Question 2 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

2 (a) The image is an electron micrograph of a human body cell.



(Source: © Science History Images/Alamy)

(i) The table gives information about parts of the human body cell.

Complete the table by giving the missing information.

(2)

Part	Name of part	Function
X		release energy from glucose
Y	nucleus	

(ii) Electron microscopes and light microscopes can both be used to view body cells.

Which of these is an advantage of using an electron microscope?

(1)

- ☒ **A** more cell structures can be seen with greater resolution
- ☒ **B** more cell structures can be seen with less resolution
- ☒ **C** fewer cell structures can be seen with greater resolution
- ☒ **D** fewer cell structures can be seen with less resolution

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P 5 8 5 6 6 A 0 4 2 8

2. Question 2 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

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(iii) The actual diameter of this human body cell before magnification is 0.05 mm.

Use information from the image to calculate the magnification of the cell.

(3)

magnification = ...

(b) Body cells produce carbon dioxide.

Describe how carbon dioxide passes from a body cell into the blood.

(2)

.....

.....

.....

.....

(Total for Question 2 = 8 marks)

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Working and answer

3. Question 3 · part a.ii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

3 (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

(2)

	Steps
	select a suitable objective lens
5	adjust focusing wheel to obtain a clear image
1	remove cells from the inside of the cheek
	place microscope slide onto the microscope stage
	smear the cell sample onto a microscope slide

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

(1)

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

(1)

.....

.....

(iv) Give two ways that the student could avoid health hazards when working with living cells.

(2)

1 .....

.....

2 .....

.....

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6

P 6 2 0 5 2 A 0 6 2 4

3. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 5 · part a.ii · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

5 The diagram shows a sperm.



(a) (i) Describe the function of a sperm.

(2)

(ii) State two other structures, apart from mitochondria, that can be seen in the sperm if it is viewed using an electron microscope.

(2)

1 .....

2 .....

(b) Explain why there are a large number of mitochondria found in a sperm.

(3)

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4. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

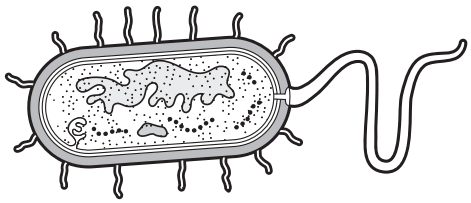
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5. Question 1 · part b.i · 3 marks

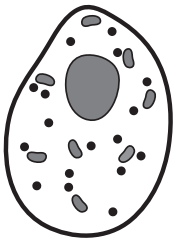
June 2024 Paper 1R

Further practice · Attempt part b.i only.

1 (a) The diagrams show a bacterial cell and a human skin cell.



Bacterial cell



Human skin cell

(i) Give **two** similarities between the bacterial cell and the human skin cell.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** differences between the bacterial cell and the human skin cell.

(2)

1 .....

.....

2 .....

.....

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5. Question 1 · part b.i · 3 marks

June 2024 Paper 1R

Further practice · Attempt part b.i only.

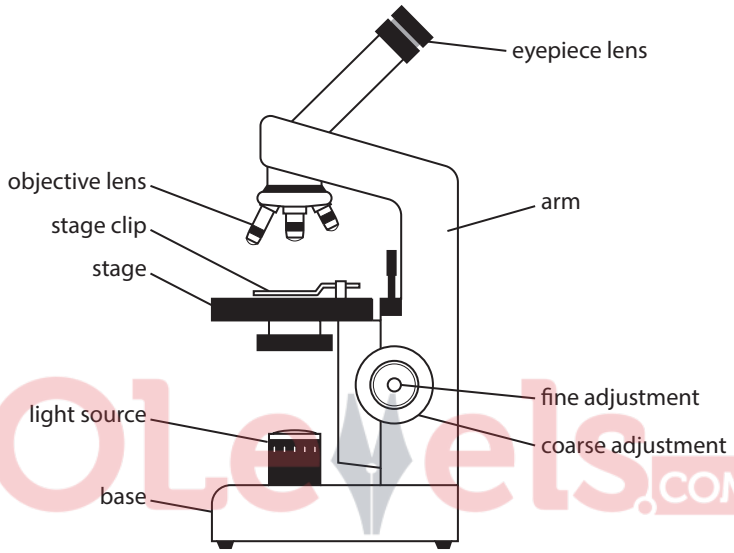
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(b) Many structures in cells can be viewed more clearly using a light microscope.

The diagram shows a light microscope.



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(i) Draw **one** straight line from each part of the light microscope to its function.

(3)

Part of microscope	Function
objective lens	holds the microscope slide in place
stage clip	moves the stage to produce a clear, focused image
fine adjustment	magnifies the sample
	keeps the microscope steady

5. Question 1 · part b.i · 3 marks

June 2024 Paper 1R

Further practice · Attempt part b.i only.

(ii) The image size of a bacterial cell viewed using an electron microscope is 5.0 cm.

The bacterial cell has been magnified 50 000 times.

Calculate the actual size of this bacterial cell in mm.

image size = actual size × magnification

(3)

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actual size of bacterial cell = ... mm

(Total for Question 1 = 10 marks)

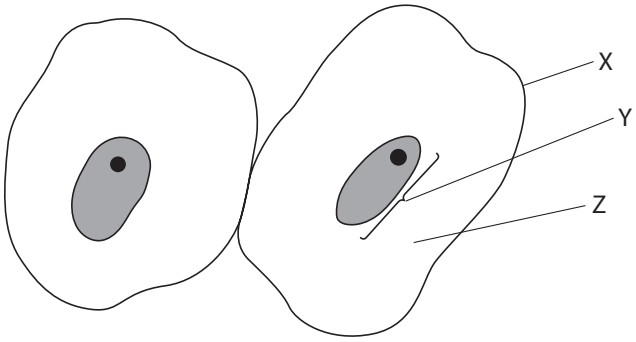
Working and answer

6. Question 7 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

7 The diagram shows two cheek cells from the inside of the mouth.



(a) Complete the table by giving the missing information.

(6)

Part	Name of part	Function of part
X		
Y		
Z		

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(b) Describe how a student would obtain and prepare these cells so the student can view them using a light microscope.

(4)

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6. Question 7 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

(c) Name two other structures inside the cells that could be seen if an electron microscope is used.

(2)

1 .....

2 .....

(Total for Question 7 = 12 marks)

Working and answer

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Practice workbook · Draw or label a human cell as seen under the microscope

Labelling given diagrams of human cells, naming lettered parts, naming a missing organelle, or drawing a labelled cheek cell.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DRAW-LABEL-ANIMAL-CELL

1. Question 2 · part a · 4 marks

November 2023 Paper 2

Worked-example retry · Attempt part a only.

2 (a) Draw a labelled diagram of a cheek cell from inside a mouth as seen under a light microscope.

(4)

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(b) (i) Explain what is meant by the term **tissue**.

(3)

(ii) State the type of tissue that contains cheek cells.

(1)

(iii) Name two other types of tissue.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

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1. Question 2 · part a · 4 marks · Working and answer

Working and answer

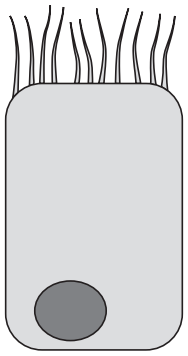
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2. Question 2 · part a.i · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

2 The diagram shows a cell from the human breathing system.



(a) (i) Add these labels to the diagram.

- cilia
- cell membrane
- nucleus

(3)

(ii) Each of the cell structures has a particular function.

The boxes give a list of structures and functions.

Draw one straight line from each structure to its function.

(3)

Structure	Function
cilia	holds the genetic code
cell membrane	site where most chemical reactions occur
nucleus	controls the entry and exit of substances
	makes proteins
	moves mucus through the trachea

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2. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

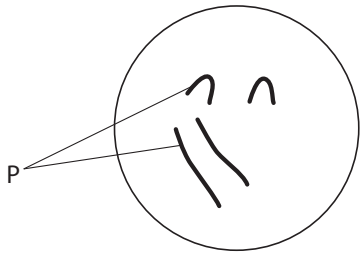
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3. Question 8 · part a.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part a.i only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

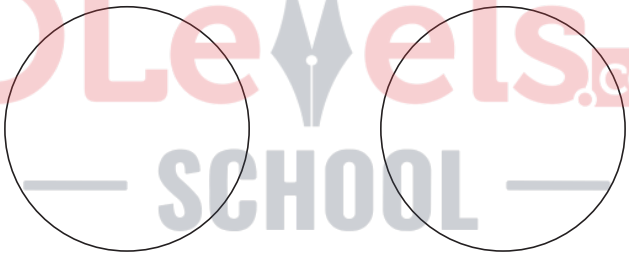
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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P 6 2 0 5 1 A 0 2 2 2 4

3. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

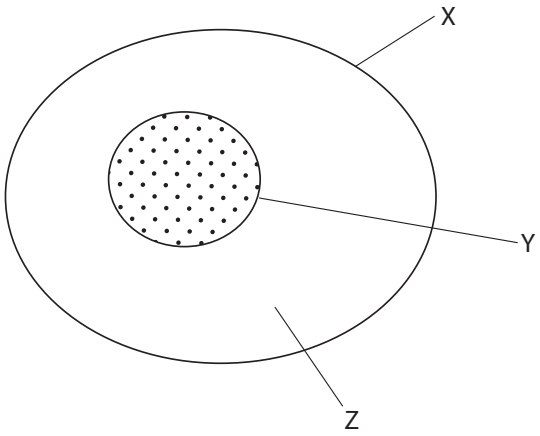
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4. Question 1 · part a.i · 3 marks

June 2022 Paper 1R

Further practice · Attempt part a.i only.

1 (a) The diagram shows a human body cell.



(i) Name the parts X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

(1)

.....

(iii) Explain how body cells are organised to form organs.

(2)

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4. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

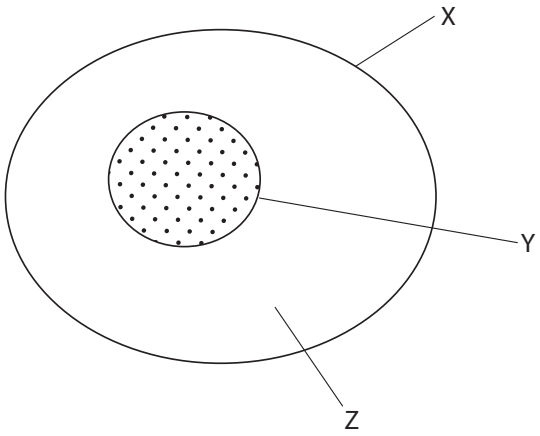
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5. Question 1 · part a.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

1 (a) The diagram shows a human body cell.



(i) Name the parts X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

(1)

.....

(iii) Explain how body cells are organised to form organs.

(2)

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P 7 0 9 4 9 A 0 2 2 4

5. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

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6. Question 6 · part a · 4 marks

June 2023 Paper 1R

Further practice · Attempt part a only.

6 The lining of the inside of the cheek is made of squamous epithelial cells.

(a) Draw a labelled diagram to show a cheek cell when seen using a light microscope.

(4)

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(b) Describe how a sample of cheek cells would be prepared to view using a light microscope.

(3)

P 7 2 5 8 6 A 0 1 4 2 0

6. Question 6 · part a · 4 marks · Working and answer

Working and answer

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Practice workbook · Name the parts of a nucleotide and the DNA backbone

Labelling phosphate, deoxyribose and base on a nucleotide diagram, naming the repeating unit, and identifying backbone or rung components.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-NAME-NUCLEOTIDE-PARTS

1. Question 1 · part c · 3 marks

June 2022 Paper 1

Worked-example retry · Attempt part c only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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P 7 0 9 7 3 A 0 2 2 4

1. Question 1 · part c · 3 marks

June 2022 Paper 1

Worked-example retry · Attempt part c only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1 .....

2 .....

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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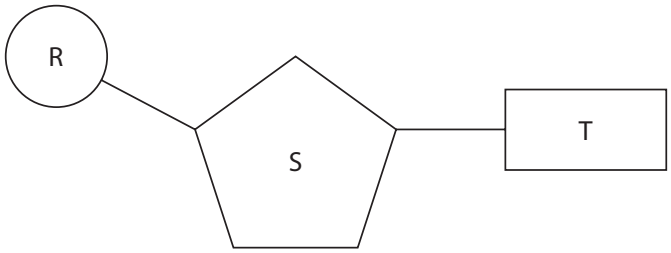
1. Question 1 · part c · 3 marks

June 2022 Paper 1

Worked-example retry · Attempt part c only.

(c) DNA is formed from molecules called nucleotides.

The diagram shows a single nucleotide.



Name the parts of the nucleotide labelled R, S and T.

(3)

R .....

S .....

T .....

(Total for Question 1 = 11 marks)

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Working and answer

2. Question 8 · part a.v · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.v only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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2. Question 8 · part a.v · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.v only.

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(v) Describe the structure of a nucleotide found in a DNA molecule.

(3)

(b) There are 20 different amino acids that can be joined to form a polypeptide chain.

The table gives some base sequences that code for five of these amino acids.

Code	Name of amino acid
AAT or AAC	asparagine
TAT or TAC	tyrosine
TTT or TTC	phenylalanine
CGT or CGG	arginine
CAT or CAC	histidine

Diagram 1 shows part of the gene that codes for a CFTR protein that functions normally.

Diagram 2 shows the same part of the gene with a mutation that causes cystic fibrosis.

AATATCATCTTTGGTGTTTCCTATGAT

Diagram 1

AATATCATCGGTGTTTCCTATGAT

Diagram 2

Describe the changes that occur in the gene and the protein that cause cystic fibrosis.

(3)

(Total for Question 8 = 14 marks)

2. Question 8 · part a.v · 3 marks · Working and answer

Working and answer

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3. Question 1 · part a.i · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.i only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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3. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 1 · part a.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.ii only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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4. Question 1 · part a.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.ii only.

(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1 .....

2 .....

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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4. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

1 (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

(3)

Cell structure	Function		
	makes proteins	**contains DNA**	**allows substances in and out of cells**
nucleus			
ribosomes			
cell membrane			

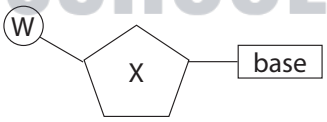
(ii) Give a structure found in bacterial cells.

(1)

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.



Name part W and part X.

(2)

W .....

X .....

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5. Question 1 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

(ii) Describe how bases hold the two chains of nucleotides together.

(3)

.....

.....

.....

.....

.....

.....

(c) The gene for the hormone insulin contains 153 bases.

Determine the number of amino acids in insulin.

(2)

number of amino acids = ...

(Total for Question 1 = 11 marks)

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5. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

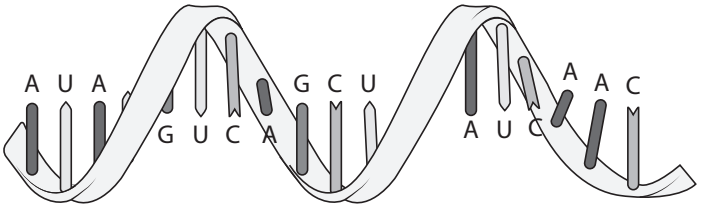
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6. Question 2 · part a.iii · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.iii only.

2 (a) The diagram shows a strand of RNA found in a body cell.



(Source: https://upload.wikimedia.org/wikipedia/commons/3/37/Difference_DNA_RNA-EN.svg)

(i) State two places where RNA is found in a cell.

(2)

1 .....

2 .....

(ii) State two differences between RNA and DNA.

(2)

1 .....

.....

2 .....

.....

(iii) RNA and DNA are both made of repeating subunits joined in a chain.

Give the name of these repeating subunits.

(1)

.....

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6. Question 2 · part a.iii · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.iii only.

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(b) Before a cell divides by mitosis, the DNA inside the cell is replicated.

When DNA is replicated, mutations can happen.

Explain how mutations can affect the final product made.

(3)

(Total for Question 2 = 8 marks)

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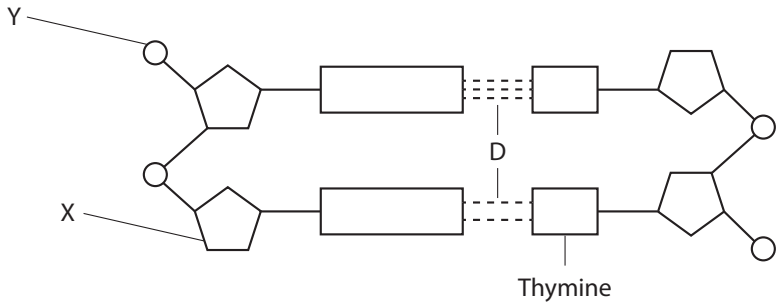
Working and answer

7. Question 2 · part a.i · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.i only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

.....

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.....

(iv) State where this nucleic acid is found in a human cell.

(1)

.....

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P 7 8 7 6 8 A 0 4 2 0

7. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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Practice workbook · Define specialised cells, tissues and organs and identify human tissues

Definitions of specialised cell, tissue and organ, naming epithelial and muscle tissues, comparing cheek and tracheal cells, and describing cardiac, smooth and skeletal muscle.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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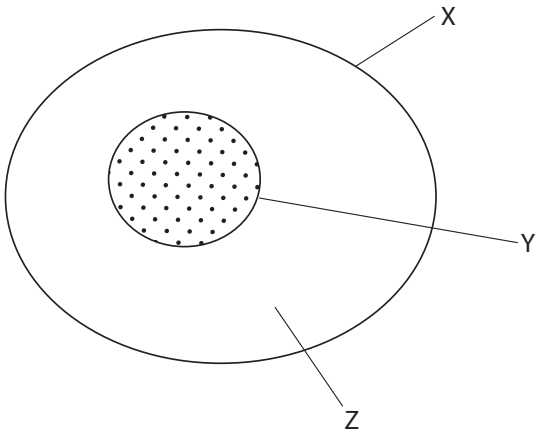
Type ID: HBI-CELLS-TISSUES-ORGANS

1. Question 1 · part a.iii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.iii only.

1 (a) The diagram shows a human body cell.



(i) Name the parts X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

(1)

.....

(iii) Explain how body cells are organised to form organs.

(2)

.....

.....

.....

.....

.....

.....

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P 7 0 9 4 9 A 0 2 2 4

1. Question 1 · part a.iii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.iii only.

(b) The boxes list three processes that transport substances in and out of cells.

Complete the diagram by drawing one straight line from each process to its correct definition.

(3)

Process	Definition
	requires energy to move molecules against a concentration gradient
osmosis	
	water moves from a high water potential to a low water potential
diffusion	
	water moves from a low water potential to a high water potential
active transport	
	molecules move down a concentration gradient

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1. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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2. Question 2 · part b.i · 3 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

2 (a) Draw a labelled diagram of a cheek cell from inside a mouth as seen under a light microscope.

(4)

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(b) (i) Explain what is meant by the term **tissue**.

(3)

(ii) State the type of tissue that contains cheek cells.

(1)

(iii) Name two other types of tissue.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

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2. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 3 · part a · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

3 (a) Describe what is meant by a specialised cell.

(2)

.....

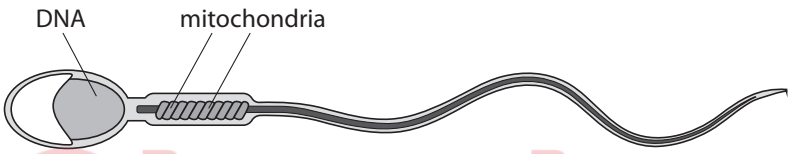
.....

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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE



(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

(2)

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

(1)

.....

.....

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P 7 2 5 8 8 A 0 8 2 4

3. Question 3 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 7 · part b.i · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.i only.

7 (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

(1)

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

(1)

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

(2)

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4. Question 7 · part b.i · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.i only.

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(ii) A motor neurone is a specialised cell.

Draw a diagram of a motor neurone.

Your diagram should include these labels.

- nucleus
- cell body
- myelin sheath

(4)

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(iii) Describe the function of a motor neurone in the nervous system.

(2)

4. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

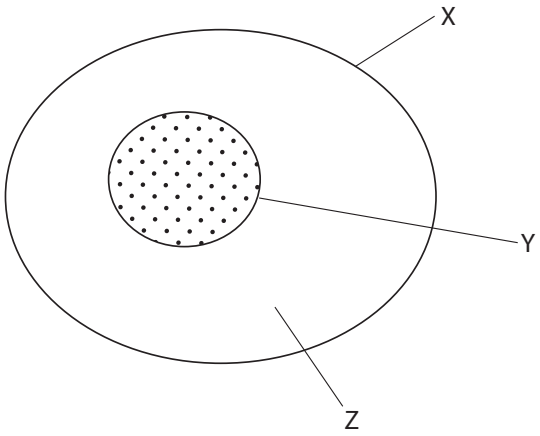
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5. Question 1 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

1 (a) The diagram shows a human body cell.



(i) Name the parts X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

(1)

.....

(iii) Explain how body cells are organised to form organs.

(2)

.....

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5. Question 1 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

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(b) The boxes list three processes that transport substances in and out of cells.

Complete the diagram by drawing one straight line from each process to its correct definition.

(3)

Process	Definition
	requires energy to move molecules against a concentration gradient
osmosis	
	water moves from a high water potential to a low water potential
diffusion	
	water moves from a low water potential to a high water potential
active transport	
	molecules move down a concentration gradient

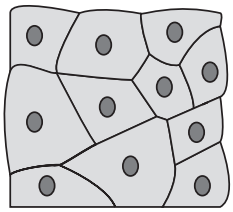
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5. Question 1 · part c.i · 2 marks

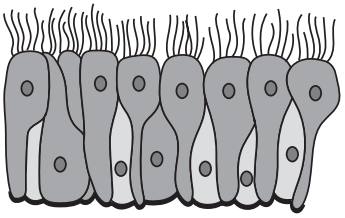
June 2022 Paper 1R

Further practice · Attempt part c.i only.

(c) (i) The diagram shows two types of epithelial tissue, A and B.



Tissue A



Tissue B

(Source: © Timonina/Shutterstock)

Give the name of each type of epithelial tissue.

(2)

A .....

B .....

(ii) An epithelial cell has an actual width of 60 micrometres (μm).

The cell is magnified 40 times.

Calculate the image width of this cell in mm.

[1 mm = 1000 μm]

(3)

image width = ... mm

(Total for Question 1 = 14 marks)

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5. Question 1 · part c.i · 2 marks · Working and answer

Working and answer

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6. Question 6 · part c · 3 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

6 The lining of the inside of the cheek is made of squamous epithelial cells.

(a) Draw a labelled diagram to show a cheek cell when seen using a light microscope.

(4)

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(b) Describe how a sample of cheek cells would be prepared to view using a light microscope.

(3)

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6. Question 6 · part c · 3 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

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(c) State three differences between a cheek cell and a cell from the lining of the trachea.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(Total for Question 6 = 10 marks)

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Working and answer

7. Question 2 · part b.ii · 1 mark

November 2023 Paper 2

Further practice · Attempt part b.ii only.

2 (a) Draw a labelled diagram of a cheek cell from inside a mouth as seen under a light microscope.

(4)

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(b) (i) Explain what is meant by the term **tissue**.

(3)

(ii) State the type of tissue that contains cheek cells.

(1)

(iii) Name two other types of tissue.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

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7. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

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8. Question 2 · part b.iii · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.iii only.

2 (a) Draw a labelled diagram of a cheek cell from inside a mouth as seen under a light microscope.

(4)

(b) (i) Explain what is meant by the term **tissue**.

(3)

(ii) State the type of tissue that contains cheek cells.

(1)

(iii) Name two other types of tissue.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

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8. Question 2 · part b.iii · 2 marks · Working and answer

Working and answer

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9. Question 5 · part a · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 dm^3 of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 dm^3 of blood contains 5.9×10^{12} red blood cells. In females the same volume of blood contains 5.5×10^{12} red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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9. Question 5 · part a · 1 mark · Working and answer

Working and answer

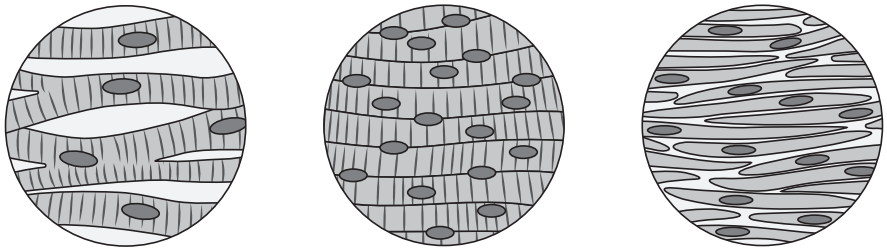
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10. Question 6 · part a.i · 6 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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10. Question 6 · part a.i · 6 marks · Working and answer

Working and answer

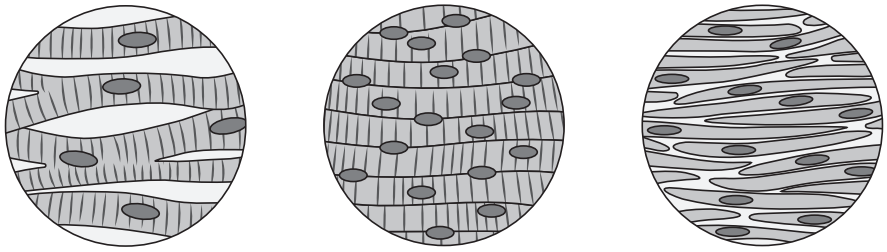
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11. Question 6 · part a.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.ii only.

6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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11. Question 6 · part a.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.ii only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

74%	75%	70%	82%	74%	74%	78%	82%	70%	82%
78%	74%	82%	78%	82%	75%	84%	72%	82%	72%

(i) Calculate the mean percentage muscle mass.

(2)

mean percentage muscle mass = ...%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

(3)

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(iii) Determine the mode of these percentage muscle masses.

(1)

mode = ...

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11. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Match human cell structures to their functions

Naming an organelle from its job, or giving the job of a named organelle, including nuclear membrane, ER and mitochondria in context.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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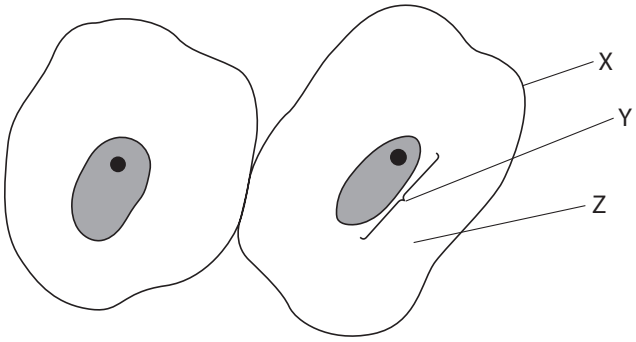
Type ID: HBI-MATCH-ORGANELLE-TO-FUNCTION

1. Question 7 · part a · 6 marks

June 2025 Paper 1R

Worked-example retry · Attempt part a only.

7 The diagram shows two cheek cells from the inside of the mouth.



(a) Complete the table by giving the missing information.

(6)

Part	Name of part	Function of part
X		
Y		
Z		

(b) Describe how a student would obtain and prepare these cells so the student can view them using a light microscope.

(4)

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1. Question 7 · part a · 6 marks · Working and answer

Working and answer

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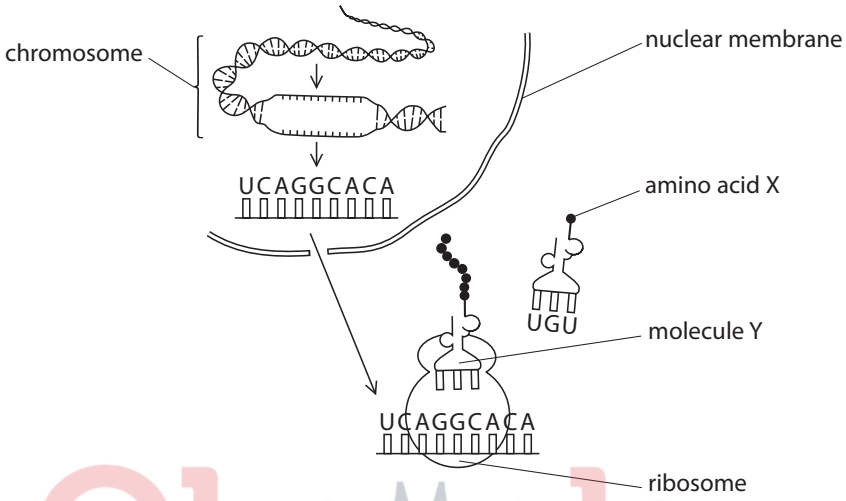
2. Question 6 · part a · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

6 Protein synthesis has a number of stages.

The diagram shows some of the stages.



The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

Base triplet of DNA	Amino acid
AGT	serine
CCG	glycine
TGT	threonine
GTA	histidine
CAA	valine
TCC	argenine
ACA	cysteine

(a) Describe the function of the nuclear membrane.

(2)

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P 6 0 1 8 6 A 0 1 6 2 8

2. Question 6 · part a · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

(1)

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

(1)

(d) Describe the process occurring on the ribosome.

(6)

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2. Question 6 · part a · 2 marks · Working and answer

Working and answer

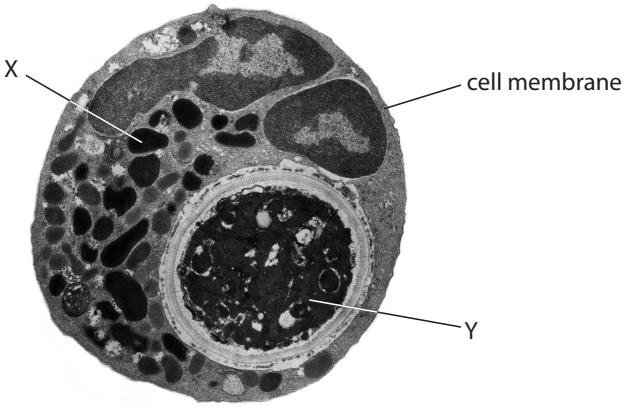
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3. Question 2 · part a.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.i only.

2 (a) The image is an electron micrograph of a human body cell.



(Source: © Science History Images/Alamy)

(i) The table gives information about parts of the human body cell.

Complete the table by giving the missing information.

(2)

Part	Name of part	Function
X		release energy from glucose
Y	nucleus	

(ii) Electron microscopes and light microscopes can both be used to view body cells.

Which of these is an advantage of using an electron microscope?

(1)

☒ **A** more cell structures can be seen with greater resolution

☒ **B** more cell structures can be seen with less resolution

☒ **C** fewer cell structures can be seen with greater resolution

☒ **D** fewer cell structures can be seen with less resolution

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3. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

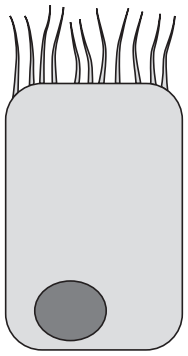
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4. Question 2 · part a.ii · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

2 The diagram shows a cell from the human breathing system.



(a) (i) Add these labels to the diagram.

- cilia
- cell membrane
- nucleus

(3)

(ii) Each of the cell structures has a particular function.

The boxes give a list of structures and functions.

Draw one straight line from each structure to its function.

(3)

Structure	Function
cilia	holds the genetic code
cell membrane	site where most chemical reactions occur
nucleus	controls the entry and exit of substances
	makes proteins
	moves mucus through the trachea

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4. Question 2 · part a.ii · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

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(b) (i) Cigarette smoking affects the function of the cell shown in the diagram.

Name a substance in cigarette smoke that causes damage to this cell.

(1)

(ii) Describe how cigarette smoking affects the cell shown in the diagram.

(2)

(Total for Question 2 = 9 marks)

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Working and answer

5. Question 8 · part a.iii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iii only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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5. Question 8 · part a.iii · 2 marks · Working and answer

Working and answer

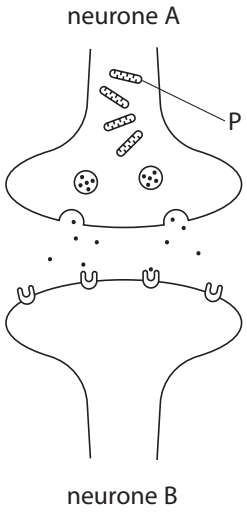
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6. Question 8 · part a.ii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

8 The diagram shows the gap between two neurones, A and B.



(a) (i) Draw an arrow on the diagram to show the direction of travel of an impulse.

(1)

(ii) Explain the function of structure P in the transmission of the impulse across the gap.

(3)

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P 6 2 0 5 4 A 0 2 2 2 4

6. Question 8 · part a.ii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

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(b) Curare is a poison used by South American Indians when hunting. Curare blocks receptors on the post synaptic membranes of neurones supplying skeletal muscles and prevents breathing.

(i) Explain how curare can stop the process of breathing.

(2)

(ii) Acetylcholinesterase is an enzyme found in the gap between two neurones.

Suggest why acetylcholinesterase inhibitors can be used to reverse the effects of curare poisoning.

(2)

(Total for Question 8 = 8 marks)

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6. Question 8 · part a.ii · 3 marks · Working and answer

Working and answer

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7. Question 2 · part a.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.ii only.

2 (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

(1)

(ii) Which of these, found in the mid-piece, release energy?

(1)

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

(3)

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7. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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8. Question 1 · part a.i · 3 marks

June 2023 Paper 2R

Further practice · Attempt part a.i only.

1 (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

(3)

Cell structure	Function		
	makes proteins	**contains DNA**	**allows substances in and out of cells**
nucleus			
ribosomes			
cell membrane			

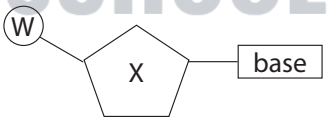
(ii) Give a structure found in bacterial cells.

(1)

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.



Name part W and part X.

(2)

W .....

X .....

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8. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

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9. Question 4 · part a.iii · 1 mark

June 2023 Paper 2

Further practice · Attempt part a.iii only.

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4 (a) (i) Complete the word equation for one type of respiration.

(2)

oxygen + ... → ..... + ...

(ii) State the type of respiration shown by the equation.

(1)

.....

(iii) In which part of the cell does this type of respiration occur?

(1)

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

(1)

.....

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9. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

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10. Question 2 · part a · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a only.

2 (a) Cells are made up of different components. Each component has a particular function.

Identify the cell component that transports substances inside the cell.

(1)

- ☒ **A** cell membrane
- ☒ **B** cytoplasm
- ☒ **C** endoplasmic reticulum
- ☒ **D** ribosomes

(b) Mitosis is a type of cell division.

A student uses a light microscope to view a cell undergoing four stages in mitosis.

The diagrams, W, X, Y and Z, show the student's observations.



W X Y Z

(Source: https://www.pinterest.co.uk/pin/812547957739062607/)

(i) Give the letters to show the correct order of the diagrams in the process of mitosis.

(1)

.....

(ii) State one function of mitosis in body cells.

(1)

.....

.....

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10. Question 2 · part a · 1 mark · Working and answer

Working and answer

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Practice workbook · Prepare cheek-cell or blood slides for a light microscope

Obtaining cheek cells or blood, smearing, staining and mounting for viewing; ordering the steps or explaining why cells were invisible.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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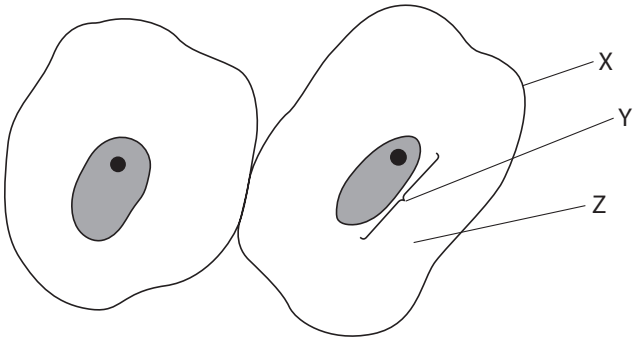
Type ID: HBI-PRAC-PREPARE-MICROSCOPE-SLIDE

1. Question 7 · part b · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part b only.

7 The diagram shows two cheek cells from the inside of the mouth.



(a) Complete the table by giving the missing information.

(6)

Part	Name of part	Function of part
X		
Y		
Z		

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(b) Describe how a student would obtain and prepare these cells so the student can view them using a light microscope.

(4)

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1. Question 7 · part b · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part b only.

(c) Name two other structures inside the cells that could be seen if an electron microscope is used.

(2)

1 .....

2 .....

(Total for Question 7 = 12 marks)

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Working and answer

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2. Question 3 · part a.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

3 (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

(2)

	Steps
	select a suitable objective lens
5	adjust focusing wheel to obtain a clear image
1	remove cells from the inside of the cheek
	place microscope slide onto the microscope stage
	smear the cell sample onto a microscope slide

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

(1)

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

(1)

.....

.....

(iv) Give two ways that the student could avoid health hazards when working with living cells.

(2)

1 .....

.....

2 .....

.....

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6

P 6 2 0 5 2 A 0 6 2 4

2. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 3 · part a.iii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a.iii only.

3 (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

(2)

	Steps
	select a suitable objective lens
5	adjust focusing wheel to obtain a clear image
1	remove cells from the inside of the cheek
	place microscope slide onto the microscope stage
	smear the cell sample onto a microscope slide

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

(1)

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

(1)

.....

.....

(iv) Give two ways that the student could avoid health hazards when working with living cells.

(2)

1 .....

.....

2 .....

.....

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3. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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4. Question 6 · part b · 3 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

6 The lining of the inside of the cheek is made of squamous epithelial cells.

(a) Draw a labelled diagram to show a cheek cell when seen using a light microscope.

(4)

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(b) Describe how a sample of cheek cells would be prepared to view using a light microscope.

(3)

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4. Question 6 · part b · 3 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

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(c) State three differences between a cheek cell and a cell from the lining of the trachea.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(Total for Question 6 = 10 marks)

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Working and answer

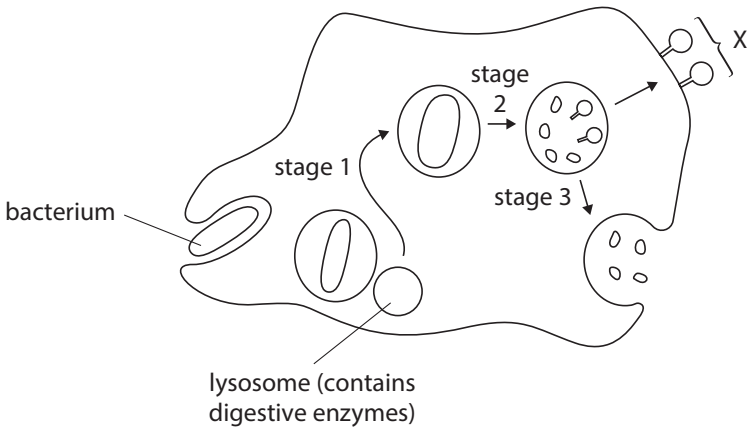
5. Question 6 · part b.i · 3 marks

June 2023 Paper 1

Further practice · Attempt part b.i only.

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6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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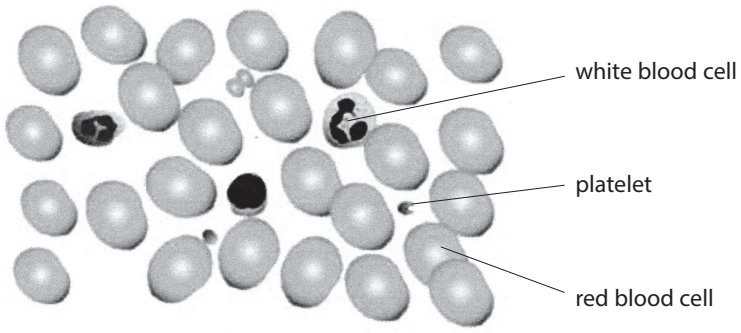
5. Question 6 · part b.i · 3 marks

June 2023 Paper 1

Further practice · Attempt part b.i only.

(b) A student observes a prepared slide of blood under a light microscope.

The student draws this diagram of what could be seen.



(i) Describe how the slide of human blood could be prepared.

(3)

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5. Question 6 · part b.i · 3 marks

June 2023 Paper 1

Further practice · Attempt part b.i only.

(ii) Discuss the accuracy of the student's diagram.

(4)

(iii) Explain why the student would be advised to use a prepared slide rather than making one from their own blood.

(2)

(Total for Question 6 = 15 marks)

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5. Question 6 · part b.i · 3 marks · Working and answer

Working and answer

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6. Question 7 · part b · 5 marks

June 2025 Paper 2

Further practice · Attempt part b only.

7 (a) Two children, M and N, are born to two sets of parents.

Child M has blood group O and child N has blood group A.

The blood groups of the two sets of parents are given in the table.

	Blood group	
	mother	**father**
1st set of parents	AB	O
2nd set of parents	B	B

(i) Explain which child belongs to which set of parents.

(4)

(ii) Calculate the probability that the next child produced by the 2nd set of parents will have blood group B.

Include a genetic diagram in your answer.

(3)

probability = ...

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6. Question 7 · part b · 5 marks

June 2025 Paper 2

Further practice · Attempt part b only.

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(b) Describe how to prepare a sample of blood to be observed using a light microscope.

(5)

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(Total for Question 7 = 12 marks)

6. Question 7 · part b · 5 marks · Working and answer

Working and answer

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Practice workbook · Compare DNA with RNA

Contrasting strands, sugar and bases of DNA and RNA, naming RNA types and locations, and using structure or base data to identify a nucleic acid.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-COMPARE-DNA-RNA

1. Question 4 · part a.iii · 3 marks

November 2023 Paper 1

Worked-example retry · Attempt part a.iii only.

4 The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

(1)

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

(1)

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

(3)

Differences	DNA	Named nucleic acid
1		
2		
3		

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1. Question 4 · part a.iii · 3 marks

November 2023 Paper 1

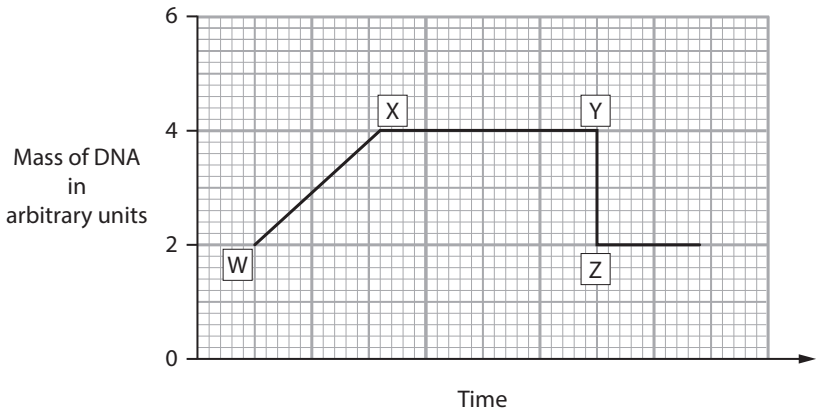
Worked-example retry · Attempt part a.iii only.

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(b) The graph shows the changes in the mass of DNA in a body cell, during mitosis.



(i) Describe the process that produces the change in the mass of DNA between points W and X.

(4)

(ii) Describe what is happening in the cell between points Y and Z.

(2)

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1. Question 4 · part a.iii · 3 marks · Working and answer

Working and answer

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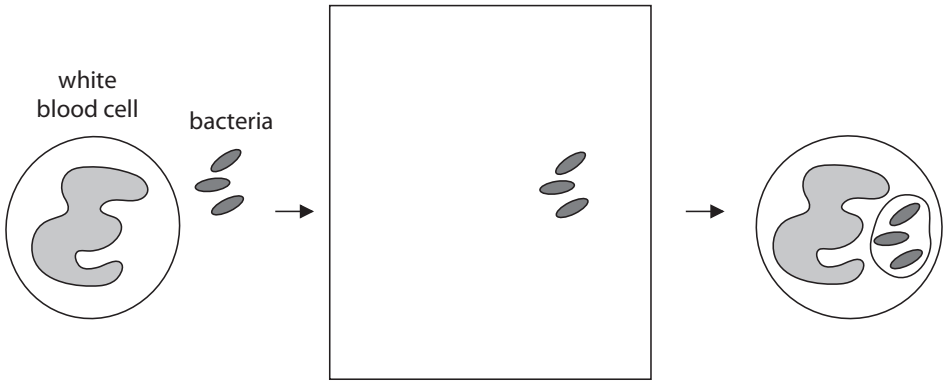
2. Question 2 · part c · 1 mark

June 2019 Paper 1

Further practice · Attempt part c only.

2 (a) Bacteria can cause disease in humans.

The diagram shows how one type of white blood cell helps to defend the body against disease.



(i) Complete the diagram by drawing the shape of the white blood cell at stage 2.

(1)

(ii) The box lists words associated with bacteria and disease.

acids	enzymes	erythrocytes	lymphocytes	phagocytes	toxins

Use words from the box to complete the sentences.

(2)

White blood cells called ..... engulf bacteria.

These white blood cells contain ..... to digest bacteria.

(iii) One way that white blood cells defend the body from disease is shown in the diagram.

State another way in which white blood cells defend the body against disease.

(1)

.....

.....

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2. Question 2 · part c · 1 mark

June 2019 Paper 1

Further practice · Attempt part c only.

(b) The table lists structures found in some cells.

Place ticks in boxes to show which structures are found in bacterial cells and which are found in human skin cells.

One row has been completed for you.

(3)

Structures	Bacterial cell	Human skin cell
nucleus		
DNA		
cytoplasm	✓	✓
cell wall		

(c) Viruses can also cause diseases in humans.

Many viruses contain RNA as their genetic material.

Which statement describes the structure of RNA?

(1)

- ☒ **A** a double-stranded helix containing the bases ATGC
- ☒ **B** a double-stranded helix containing the bases AUGC
- ☒ **C** a single-stranded helix containing the bases ATGC
- ☒ **D** a single-stranded helix containing the bases AUGC

(Total for Question 2 = 8 marks)

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Working and answer

3. Question 2 · part b · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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3. Question 2 · part b · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b only.

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(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

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(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

3. Question 2 · part b · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b only.

(iv) Suggest why donated CWB should never be frozen (line 17).

(2)

(v) Explain how the blood of people who have recovered from Ebola is likely to be effective in treating the disease.

(2)

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(b) The nitrogenous base composition of another virus was found to be adenine 13%, guanine 26%, uracil 20% and cytosine 41%.

Explain how these figures prove that this is a virus with a single strand of RNA.

(3)

(Total for Question 2 = 14 marks)

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3. Question 2 · part b · 3 marks · Working and answer

Working and answer

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4. Question 1 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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4. Question 1 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1 .....

2 .....

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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4. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 1 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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5. Question 1 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1 .....

2 .....

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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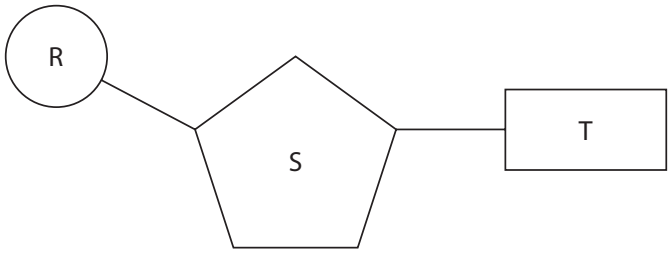
5. Question 1 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

(c) DNA is formed from molecules called nucleotides.

The diagram shows a single nucleotide.



Name the parts of the nucleotide labelled R, S and T.

(3)

R .....

S .....

T .....

(Total for Question 1 = 11 marks)

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DO NOT

Working and answer

6. Question 4 · part a.ii · 1 mark

November 2023 Paper 1

Further practice · Attempt part a.ii only.

4 The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

(1)

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

(1)

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

(3)

Differences	DNA	Named nucleic acid
1		
2		
3		

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6. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

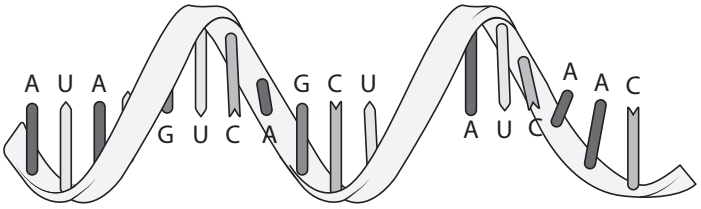
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7. Question 2 · part a.i · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.i only.

2 (a) The diagram shows a strand of RNA found in a body cell.



(Source: https://upload.wikimedia.org/wikipedia/commons/3/37/Difference_DNA_RNA-EN.svg)

(i) State two places where RNA is found in a cell.

(2)

1 .....

2 .....

(ii) State two differences between RNA and DNA.

(2)

1 .....

.....

2 .....

.....

(iii) RNA and DNA are both made of repeating subunits joined in a chain.

Give the name of these repeating subunits.

(1)

.....

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P 7 4 6 3 0 A 0 6 2 4

7. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

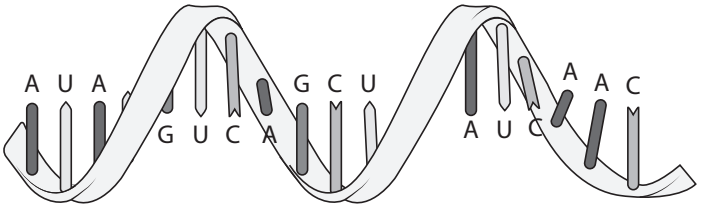
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8. Question 2 · part a.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

2 (a) The diagram shows a strand of RNA found in a body cell.



(Source: https://upload.wikimedia.org/wikipedia/commons/3/37/Difference_DNA_RNA-EN.svg)

(i) State two places where RNA is found in a cell.

(2)

1 .....

2 .....

(ii) State two differences between RNA and DNA.

(2)

1 .....

.....

2 .....

.....

(iii) RNA and DNA are both made of repeating subunits joined in a chain.

Give the name of these repeating subunits.

(1)

.....

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P 7 4 6 3 0 A 0 6 2 4

8. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

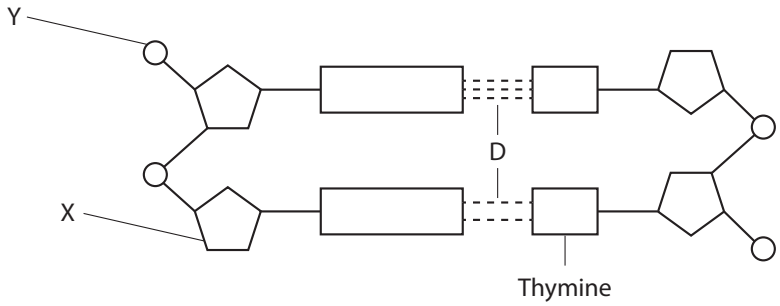
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9. Question 2 · part a.ii · 1 mark

June 2025 Paper 1

Further practice · Attempt part a.ii only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

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(iv) State where this nucleic acid is found in a human cell.

(1)

.....

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P 7 8 7 6 8 A 0 4 2 0

9. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

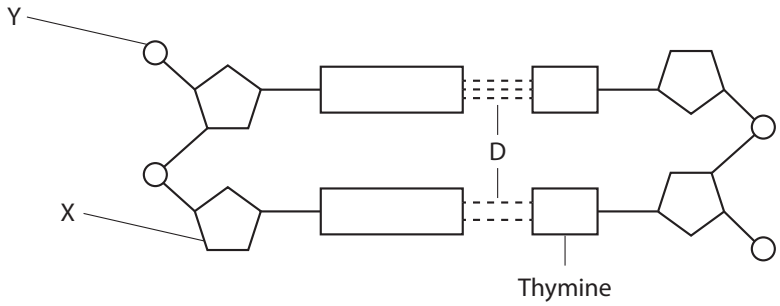
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10. Question 2 · part a.iii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.iii only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

.....

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.....

.....

(iv) State where this nucleic acid is found in a human cell.

(1)

.....

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P 7 8 7 6 8 A 0 4 2 0

10. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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Practice workbook · Describe DNA's double helix, base pairing and replication

Two strands linked by complementary bases and hydrogen bonds, deducing bases or proportions, and copying DNA with DNA polymerase before division.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DNA-STRUCTURE-AND-REPLICATION

1. Question 3 · part c · 4 marks

June 2024 Paper 1

Worked-example retry · Attempt part c only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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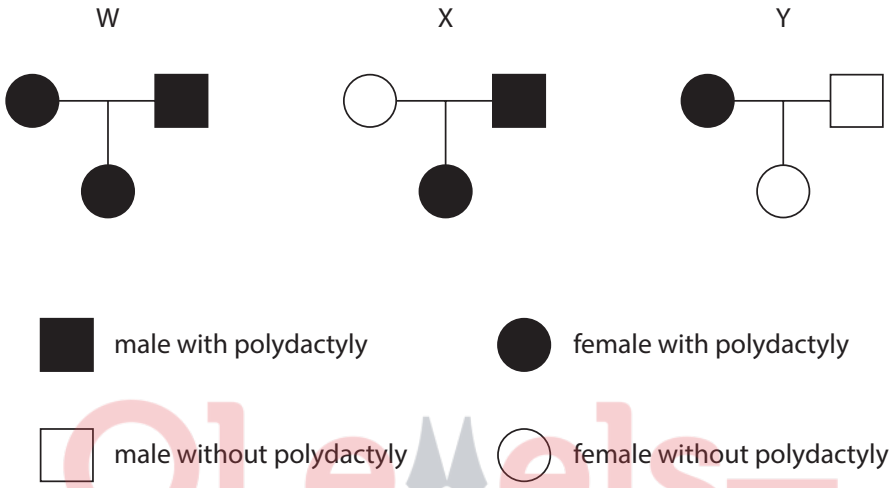
1. Question 3 · part c · 4 marks

June 2024 Paper 1

Worked-example retry · Attempt part c only.

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(ii) The family pedigree diagrams show the inheritance of polydactyly.



Which of the diagrams, W, X or Y, gives the family pedigree for parents R and S?

(1)

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(c) A gene is a short section of DNA.

Describe the structure of DNA.

(4)

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(Total for Question 3 = 9 marks)

1. Question 3 · part c · 4 marks · Working and answer

Working and answer

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2. Question 1 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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P 7 0 9 7 3 A 0 2 2 4

2. Question 1 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1

2

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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2. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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3. Question 1 · part b.ii · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

1 (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

(3)

Cell structure	Function		
	makes proteins	**contains DNA**	**allows substances in and out of cells**
nucleus			
ribosomes			
cell membrane			

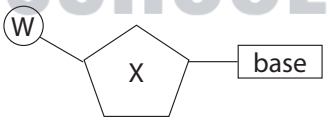
(ii) Give a structure found in bacterial cells.

(1)

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.



Name part W and part X.

(2)

W .....

X .....

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DO NOT WRITE IN THIS AREA

P 7 2 5 8 8 A 0 2 2 4

3. Question 1 · part b.ii · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Describe how bases hold the two chains of nucleotides together.

(3)

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.....

(c) The gene for the hormone insulin contains 153 bases.

Determine the number of amino acids in insulin.

(2)

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number of amino acids = ...

(Total for Question 1 = 11 marks)

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3. Question 1 · part b.ii · 3 marks · Working and answer

Working and answer

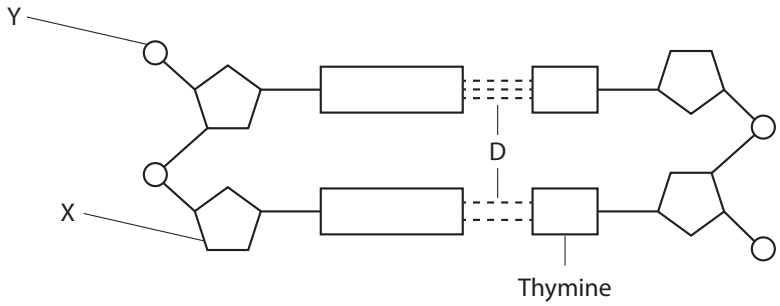
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4. Question 2 · part b.i · 1 mark

June 2025 Paper 1

Further practice · Attempt part b.i only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

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(iv) State where this nucleic acid is found in a human cell.

(1)

.....

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P 7 8 7 6 8 A 0 4 2 0

4. Question 2 · part b.i · 1 mark

June 2025 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) State what is labelled D in the diagram.

(1)

(ii) Describe the importance of D in maintaining the structure of the nucleic acid.

(3)

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(Total for Question 2 = 11 marks)

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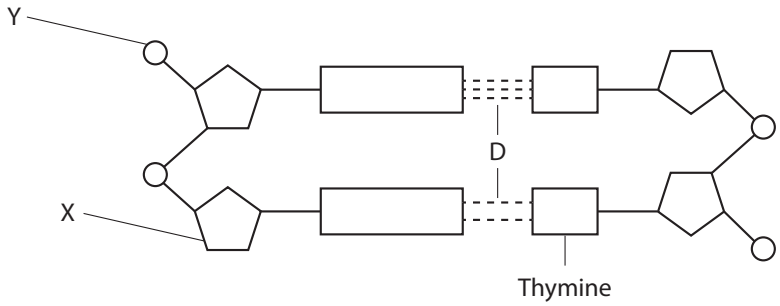
Working and answer

5. Question 2 · part b.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part b.ii only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

.....

.....

.....

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.....

.....

(iv) State where this nucleic acid is found in a human cell.

(1)

.....

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P 7 8 7 6 8 A 0 4 2 0

5. Question 2 · part b.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part b.ii only.

(b) (i) State what is labelled D in the diagram.

(1)

.....

(ii) Describe the importance of D in maintaining the structure of the nucleic acid.

(3)

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(Total for Question 2 = 11 marks)

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5. Question 2 · part b.ii · 3 marks · Working and answer

Working and answer

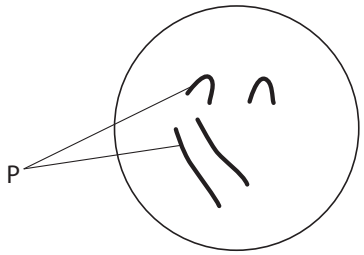
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6. Question 8 · part c.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part c.i only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

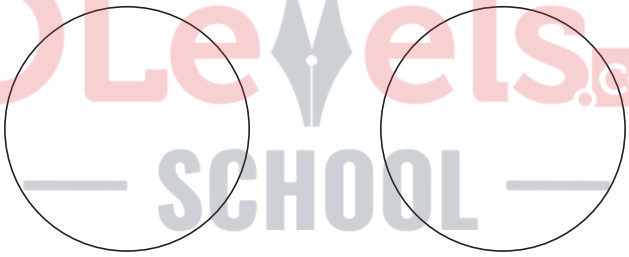
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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6. Question 8 · part c.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part c.i only.

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(c) The DNA of the nucleus is analysed.

The table shows the results.

Base	X	adenine	Y	Z
Pecentage base in DNA (%)	30	20	30	20

(i) Identify base Z.

(1)

(ii) Give the possible identities of base X.

(1)

(d) Meiosis is another type of cell division.

Explain why meiosis is important in humans.

(4)

(Total for Question 8 = 10 marks)

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6. Question 8 · part c.i · 1 mark · Working and answer

Working and answer

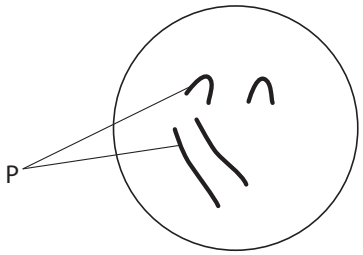
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7. Question 8 · part c.ii · 1 mark

November 2020 Paper 1

Further practice · Attempt part c.ii only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

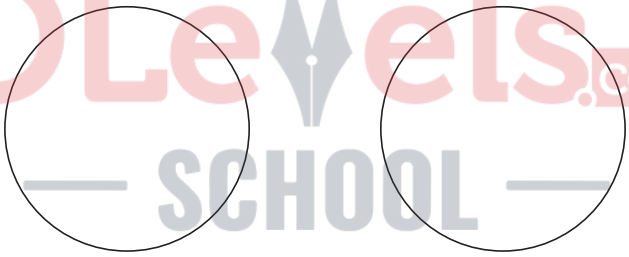
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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7. Question 8 · part c.ii · 1 mark

November 2020 Paper 1

Further practice · Attempt part c.ii only.

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(c) The DNA of the nucleus is analysed.

The table shows the results.

Base	X	adenine	Y	Z
Pecentage base in DNA (%)	30	20	30	20

(i) Identify base Z.

(1)

(ii) Give the possible identities of base X.

(1)

(d) Meiosis is another type of cell division.

Explain why meiosis is important in humans.

(4)

(Total for Question 8 = 10 marks)

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7. Question 8 · part c.ii · 1 mark · Working and answer

Working and answer

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8. Question 6 · part b · 2 marks

November 2020 Paper 1R

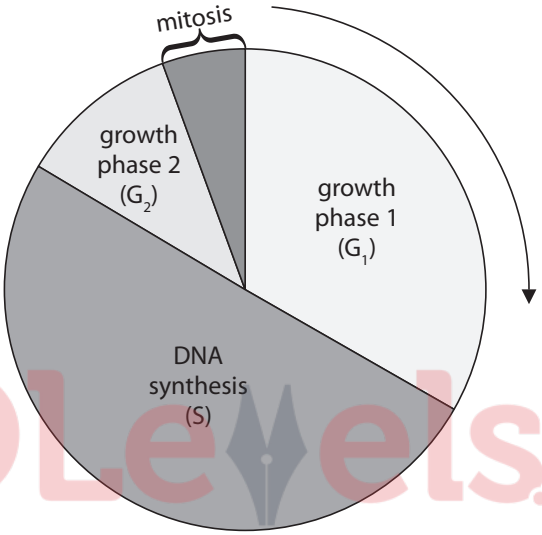
Further practice · Attempt part b only.

6 (a) Cells in the body go through several cell cycles in their lifetime.

Each cell cycle is divided into different stages.

Different events take place in the cell during each stage.

The diagram shows one cell cycle of a body cell.



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(i) One cell cycle takes 24 hours to complete.

The total time spent in G_1 and G_2 is ten hours.

Calculate the time spent in mitosis.

(2)

time = ... hours

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8. Question 6 · part b · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

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(ii) Each of the phases of the cell cycle prepare the cell for mitosis.

Suggest the events that take place in the growth phases of the cell cycle.

(2)

(iii) The cell cycle completes by producing two new body cells.

Describe the genetic features of the two cells produced by mitosis.

(2)

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(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

(3)

Stage	Events
metaphase	
anaphase	
telophase	

8. Question 6 · part b · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

(b) Describe the role of DNA polymerase in the synthesis of new DNA molecules.

(2)

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(Total for Question 6 = 11 marks)

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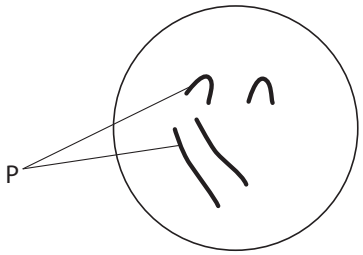
Working and answer

9. Question 8 · part b · 1 mark

November 2020 Paper 1

Further practice · Attempt part b only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

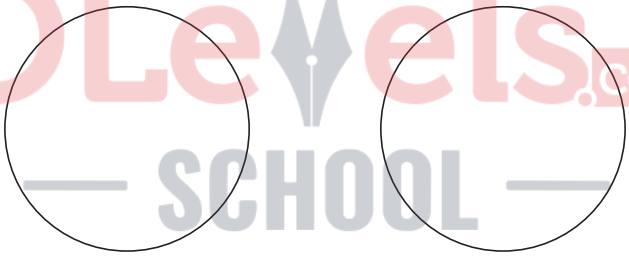
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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9. Question 8 · part b · 1 mark

November 2020 Paper 1

Further practice · Attempt part b only.

(c) The DNA of the nucleus is analysed.

The table shows the results.

Base	X	adenine	Y	Z
Pecentage base in DNA (%)	30	20	30	20

(i) Identify base Z.

(1)

(ii) Give the possible identities of base X.

(1)

(d) Meiosis is another type of cell division.

Explain why meiosis is important in humans.

(4)

(Total for Question 8 = 10 marks)

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9. Question 8 · part b · 1 mark · Working and answer

Working and answer

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10. Question 4 · part b.i · 4 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

4 The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

(1)

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

(1)

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

(3)

Differences	DNA	Named nucleic acid
1		
2		
3		

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10. Question 4 · part b.i · 4 marks

November 2023 Paper 1

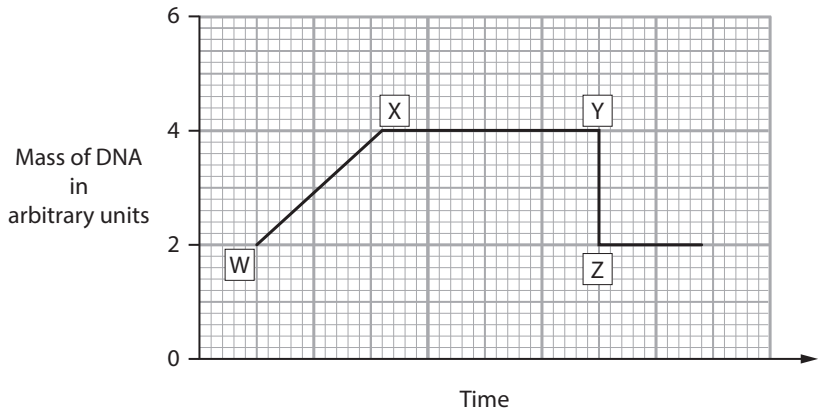
Further practice · Attempt part b.i only.

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(b) The graph shows the changes in the mass of DNA in a body cell, during mitosis.



(i) Describe the process that produces the change in the mass of DNA between points W and X.

(4)

(ii) Describe what is happening in the cell between points Y and Z.

(2)

10. Question 4 · part b.i · 4 marks · Working and answer

Working and answer

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Practice workbook · Describe the products and purposes of mitosis

What mitosis produces, drawing daughter cells or nuclei, cell cycle growth and division phases, uses of mitosis and contrast with meiosis.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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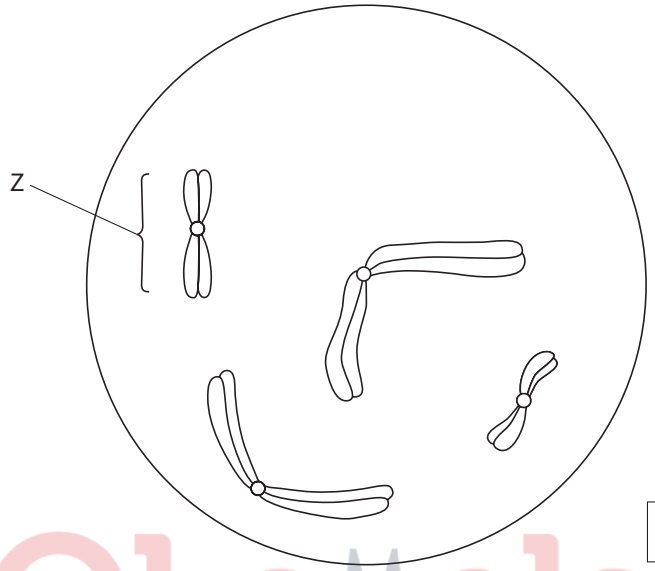
Type ID: HBI-DESCRIBE-MITOSIS-PRODUCTS-AND-USES

1. Question 7 · part c · 2 marks

November 2021 Paper 1

Worked-example retry · Attempt part c only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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P 6 7 0 6 0 R R A 0 1 6 2 0

1. Question 7 · part c · 2 marks

November 2021 Paper 1

Worked-example retry · Attempt part c only.

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(b) (i) Which one of these molecules would be found in structure Z?

(1)

- ☒ **A** DNA
- ☒ **B** mRNA
- ☒ **C** tRNA
- ☒ **D** ribose sugar

(ii) Calculate the length of structure Z in micrometres.

[1 micrometre = 1×10^{-6} m]

(3)

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length of structure Z = ... micrometres

(c) Describe what is produced when a cell undergoes mitosis.

(2)

.....

.....

.....

.....

(Total for Question 7 = 11 marks)

1. Question 7 · part c · 2 marks · Working and answer

Working and answer

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2. Question 6 · part a.ii · 2 marks

November 2020 Paper 1R

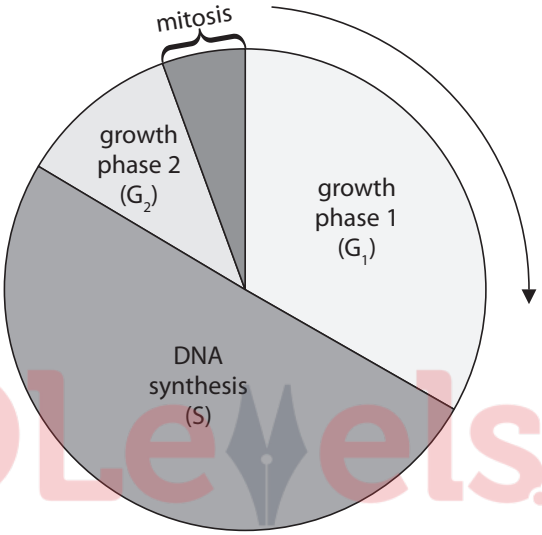
Further practice · Attempt part a.ii only.

6 (a) Cells in the body go through several cell cycles in their lifetime.

Each cell cycle is divided into different stages.

Different events take place in the cell during each stage.

The diagram shows one cell cycle of a body cell.



(i) One cell cycle takes 24 hours to complete.

The total time spent in G_1 and G_2 is ten hours.

Calculate the time spent in mitosis.

(2)

time = ... hours

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P 6 2 0 5 2 A 0 1 2 2 4

2. Question 6 · part a.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

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(ii) Each of the phases of the cell cycle prepare the cell for mitosis.

Suggest the events that take place in the growth phases of the cell cycle.

(2)

(iii) The cell cycle completes by producing two new body cells.

Describe the genetic features of the two cells produced by mitosis.

(2)

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(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

(3)

Stage	Events
metaphase	
anaphase	
telophase	

2. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part a.iii · 2 marks

November 2020 Paper 1R

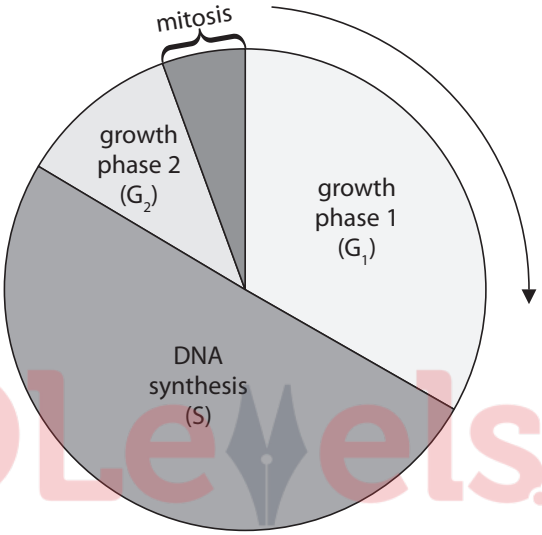
Further practice · Attempt part a.iii only.

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(2)

time = ... hours

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3. Question 6 · part a.iii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iii only.

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Suggest the events that take place in the growth phases of the cell cycle.

(2)

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Describe the genetic features of the two cells produced by mitosis.

(2)

(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

(3)

Stage	Events
metaphase	
anaphase	
telophase	

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3. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

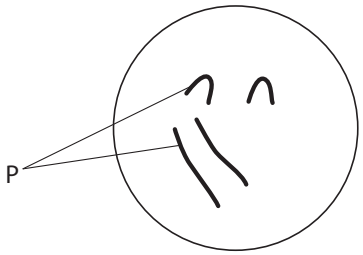
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4. Question 8 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

8 (a) The diagram shows the nucleus of a cell.



(i) Name the structures labelled P.

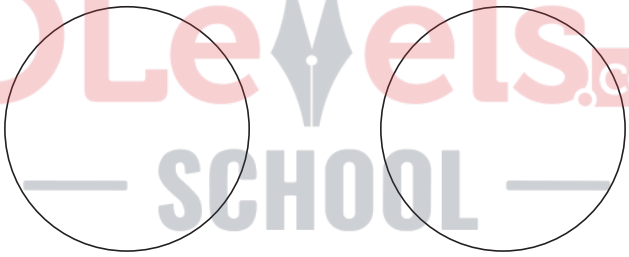
(1)

.....

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

(2)



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

(1)

Stage	Number
complementary bases pair up	
DNA double helix unwinds	
strands separate	
two DNA helices form	

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4. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

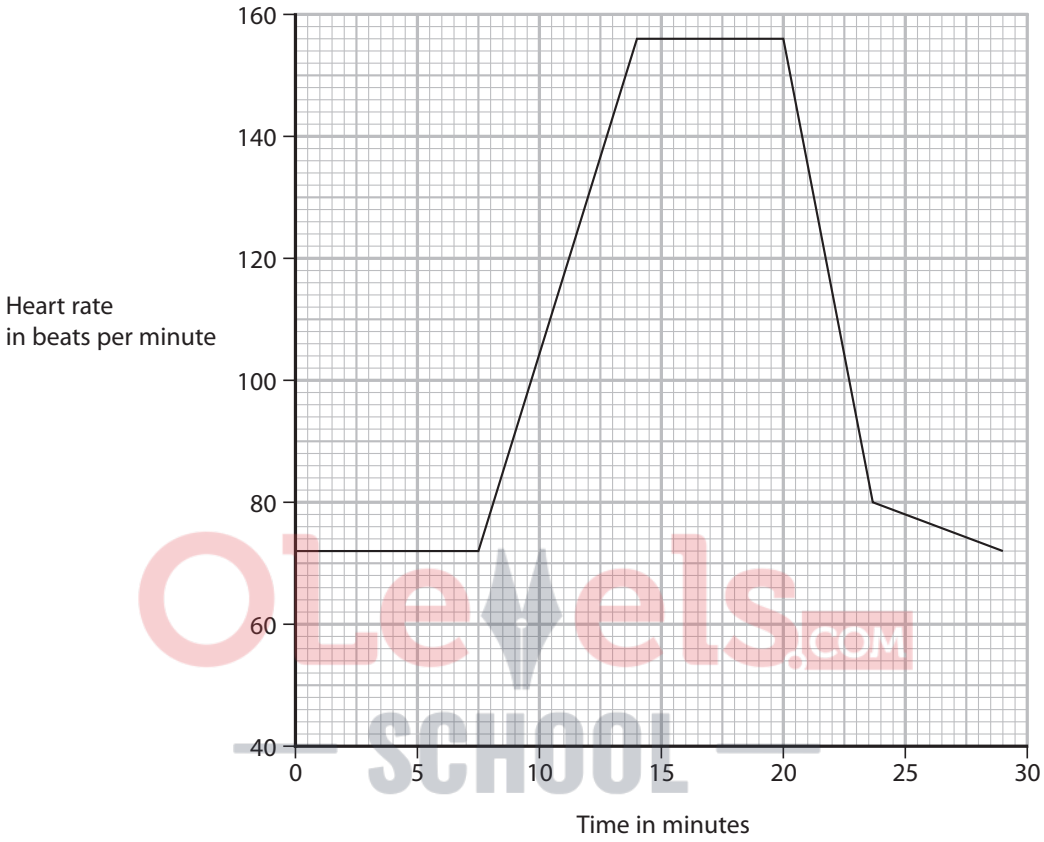
5. Question 2 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

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2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

5. Question 2 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

(iii) Complete the passage about what happens in the body during exercise using the correct words.

(4)

During exercise ..... gas moves from the lungs into the blood by a process called ..... . This gas is needed by cells for ..... respiration which releases energy. Also during exercise, the amount of urine produced ..... as water is lost in sweat.

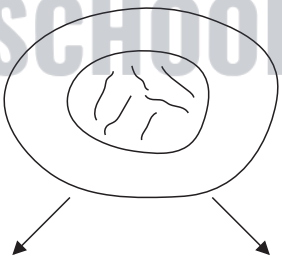
(b) (i) During exercise, body cells get damaged.

Body cells are replaced by a process called mitosis.

The diagram represents one body cell but shows only 6 of the 46 chromosomes.

Complete the diagram to show the new body cells produced from this cell by mitosis.

(2)



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P 7 0 9 7 3 A 0 6 2 4

5. Question 2 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

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(ii) Mitosis takes place in four stages.

Which of these gives the stage of mitosis where chromosomes align at the equator of the cell?

(1)

- ☒ **A** anaphase
- ☒ **B** metaphase
- ☒ **C** prophase
- ☒ **D** telophase

(Total for Question 2 = 10 marks)

Working and answer

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6. Question 4 · part b.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

4 The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

(1)

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

(1)

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

(3)

Differences	DNA	Named nucleic acid
1		
2		
3		

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6. Question 4 · part b.ii · 2 marks

November 2023 Paper 1

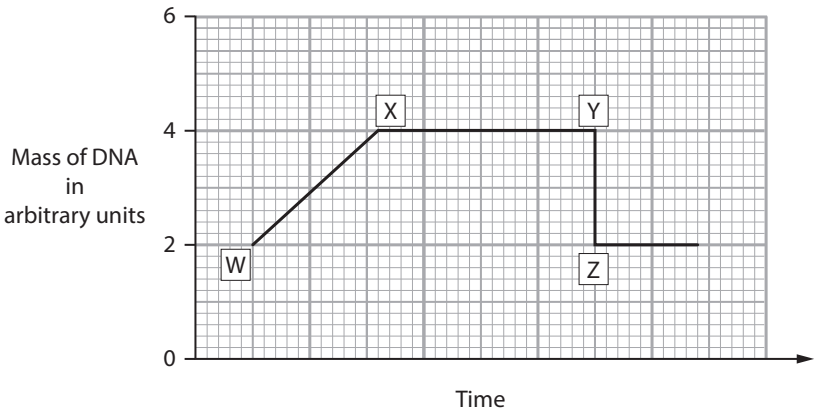
Further practice · Attempt part b.ii only.

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(b) The graph shows the changes in the mass of DNA in a body cell, during mitosis.



(i) Describe the process that produces the change in the mass of DNA between points W and X.

(4)

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(ii) Describe what is happening in the cell between points Y and Z.

(2)

6. Question 4 · part b.ii · 2 marks

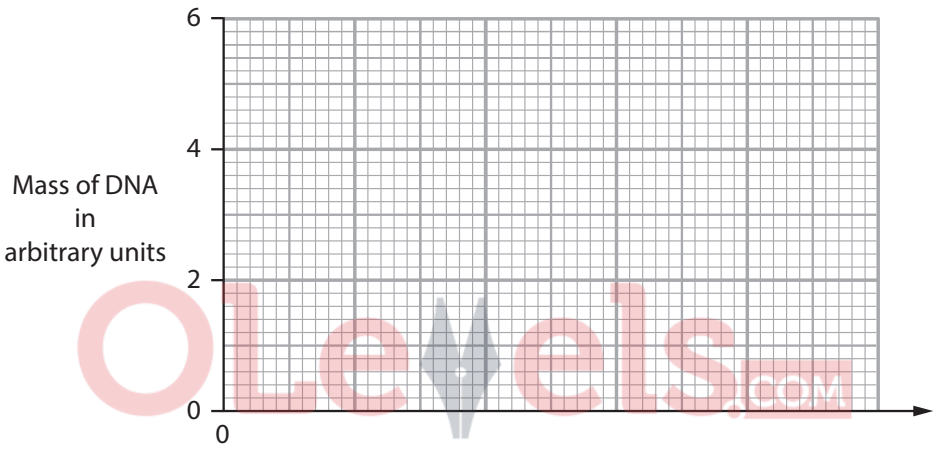
November 2023 Paper 1

Further practice · Attempt part b.ii only.

(iii) The production of gametes involves the process of meiosis.

On the grid draw a line to show the changes in the mass of DNA during meiosis.

(3)



(Total for Question 4 = 14 marks)

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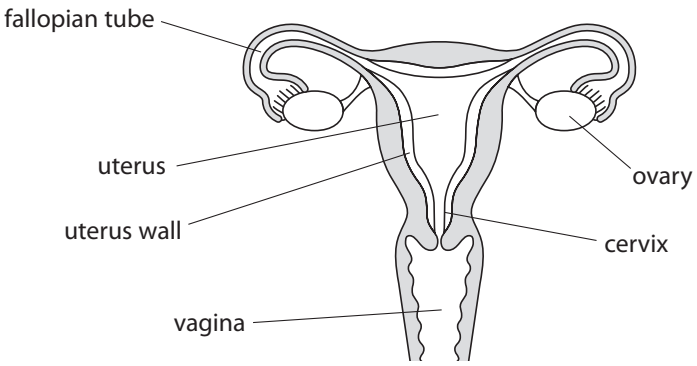
Working and answer

7. Question 2 · part b.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

2 (a) The diagram shows the female reproductive system.



(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

(3)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			
ovary			
vagina			

(ii) Describe the action of the muscles in the uterus wall during labour.

(2)

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7. Question 2 · part b.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

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(iii) Fertilisation takes place in the female reproductive system.

Which of these describes fertilisation?

(1)

- ☒ **A** a sperm cell fuses with an egg cell to form a haploid cell
- ☒ **B** a sperm cell fuses with an egg cell to form an embryo
- ☒ **C** a sperm cell fuses with an egg cell to form a zygote
- ☒ **D** a sperm cell fuses with an egg cell to form a baby

(iv) Before fertilisation can take place, ovulation needs to occur.

Name the hormone that causes ovulation.

(1)

.....

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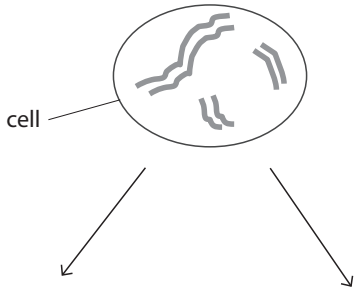
7. Question 2 · part b.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

(b) Following fertilisation and during pregnancy, cell division by mitosis takes place.

The diagram shows a cell with only three pairs of chromosomes instead of a full set.



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(i) Complete the diagram to show the cells formed when this cell divides by mitosis.

(2)

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7. Question 2 · part b.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

(ii) In humans, cell division occurs by mitosis or meiosis.

Draw one straight line from each type of cell division to the cells produced.

(2)

Type of cell division	Cells produced
	two cells, each with 23 chromosomes
mitosis	four cells, each with 46 chromosomes
meiosis	two cells, each with 46 chromosomes
	four cells, each with 23 chromosomes

(Total for Question 2 = 11 marks)

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7. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

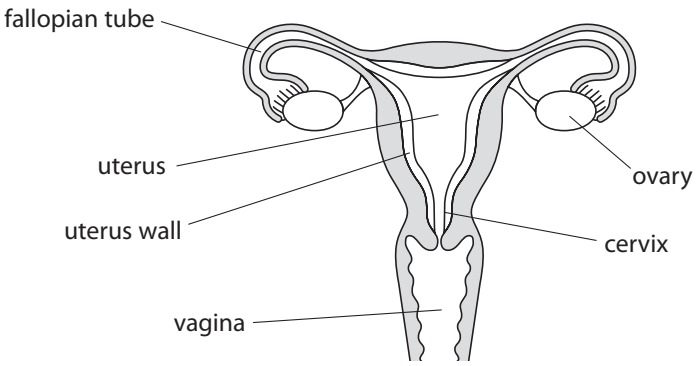
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8. Question 2 · part b.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

2 (a) The diagram shows the female reproductive system.



(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

(3)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			
ovary			
vagina			

(ii) Describe the action of the muscles in the uterus wall during labour.

(2)

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8. Question 2 · part b.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Fertilisation takes place in the female reproductive system.

Which of these describes fertilisation?

(1)

- ☒ **A** a sperm cell fuses with an egg cell to form a haploid cell
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- ☒ **D** a sperm cell fuses with an egg cell to form a baby

(iv) Before fertilisation can take place, ovulation needs to occur.

Name the hormone that causes ovulation.

(1)

.....

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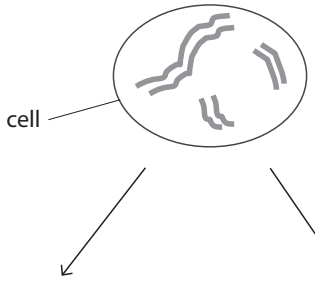
8. Question 2 · part b.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

(b) Following fertilisation and during pregnancy, cell division by mitosis takes place.

The diagram shows a cell with only three pairs of chromosomes instead of a full set.



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(i) Complete the diagram to show the cells formed when this cell divides by mitosis.

(2)

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8. Question 2 · part b.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) In humans, cell division occurs by mitosis or meiosis.

Draw one straight line from each type of cell division to the cells produced.

(2)

Type of cell division	Cells produced
	two cells, each with 23 chromosomes
mitosis	four cells, each with 46 chromosomes
meiosis	two cells, each with 46 chromosomes
	four cells, each with 23 chromosomes

(Total for Question 2 = 11 marks)

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8. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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9. Question 2 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

2 (a) Cells are made up of different components. Each component has a particular function.

Identify the cell component that transports substances inside the cell.

(1)

- ☒ **A** cell membrane
- ☒ **B** cytoplasm
- ☒ **C** endoplasmic reticulum
- ☒ **D** ribosomes

(b) Mitosis is a type of cell division.

A student uses a light microscope to view a cell undergoing four stages in mitosis.

The diagrams, W, X, Y and Z, show the student's observations.



W X Y Z

(Source: https://www.pinterest.co.uk/pin/812547957739062607/)

(i) Give the letters to show the correct order of the diagrams in the process of mitosis.

(1)

.....

(ii) State one function of mitosis in body cells.

(1)

.....

.....

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9. Question 2 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

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(c) Meiosis is another type of cell division.

Give two differences between the cells produced by meiosis and the cells produced by mitosis.

(2)

1 .....

.....

2 .....

.....

(Total for Question 2 = 5 marks)

Working and answer

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Practice workbook · Describe transcription and translation in human cells

Describing how a cell makes protein from a gene, and applying the sequence to switched-off genes, low amino acid supply or viral proteins.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DESCRIBE-PROTEIN-SYNTHESIS

1. Question 8 · part a.ii · 5 marks

June 2023 Paper 2R

Worked-example retry · Attempt part a.ii only.

8 (a) (i) Describe the structure of a virus.

(2)

(ii) Viruses attach to cells of various body tissues.

The virus uses the host cell's genetic material to produce new viral proteins.

Describe how the host cell carries out protein synthesis to produce new viral proteins.

(5)

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1. Question 8 · part a.ii · 5 marks

June 2023 Paper 2R

Worked-example retry · Attempt part a.ii only.

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(b) In 2010, AIDS-related illnesses caused the death of 1.4 million people.

In 2021, AIDS-related illnesses caused the death of 650 000 people.

(i) Name the microorganism that causes AIDS.

(1)

.....

(ii) Calculate the percentage decrease in the number of people dying from AIDS-related illnesses between 2010 and 2020.

(2)

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percentage decrease = ... %

(Total for Question 8 = 10 marks)

1. Question 8 · part a.ii · 5 marks · Working and answer

Working and answer

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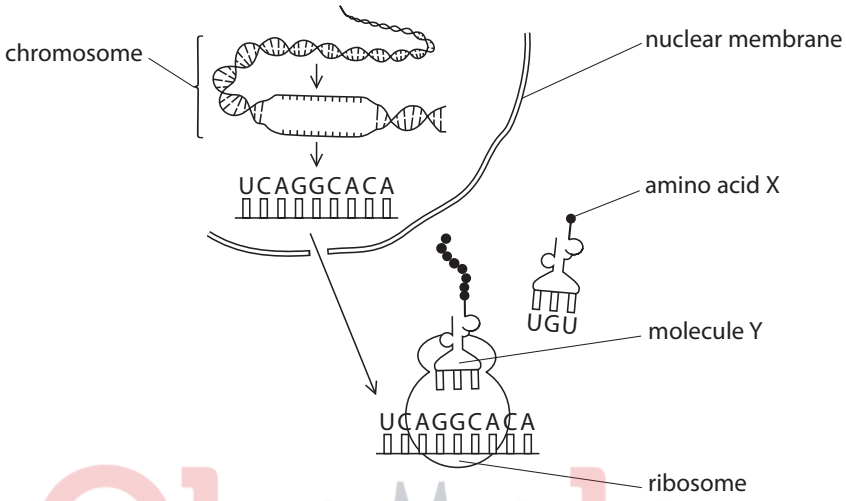
2. Question 6 · part d · 6 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

6 Protein synthesis has a number of stages.

The diagram shows some of the stages.



The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

Base triplet of DNA	Amino acid
AGT	serine
CCG	glycine
TGT	threonine
GTA	histidine
CAA	valine
TCC	argenine
ACA	cysteine

(a) Describe the function of the nuclear membrane.

(2)

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2. Question 6 · part d · 6 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

(1)

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

(1)

(d) Describe the process occurring on the ribosome.

(6)

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2. Question 6 · part d · 6 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

(e) Describe how a mutation will affect a chromosome.

(2)

(Total for Question 6 = 12 marks)

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Working and answer

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3. Question 9 · part a.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

9 (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

(3)

(ii) Explain why people with PKU are advised to follow a low-protein diet.

(3)

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

(3)

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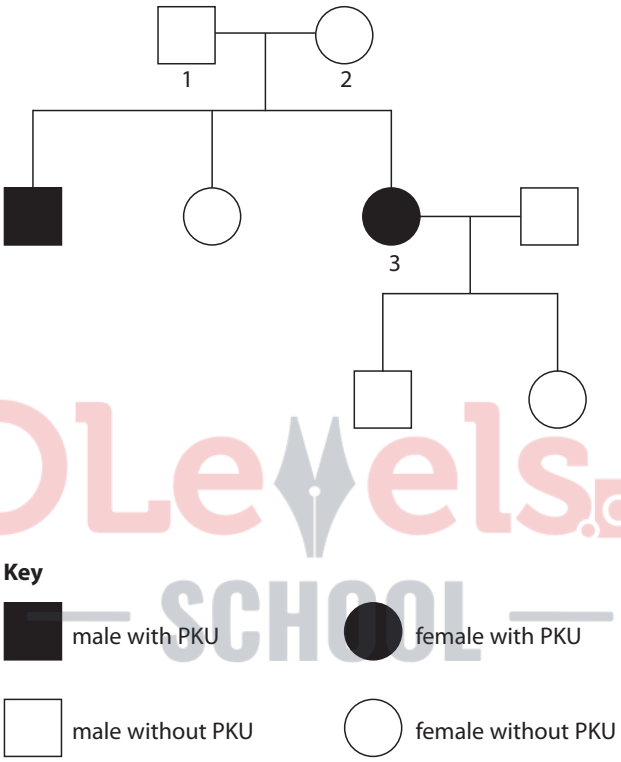
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3. Question 9 · part a.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

(b) The diagram shows how PKU is inherited in a family.



(i) The allele for PKU can be represented by n. The normal allele can be represented by N.

Give the genotype of person 1.

(1)

(ii) Determine the probability that person 2 has one allele for PKU.

(1)

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3. Question 9 · part a.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 5 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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P 7 0 9 5 0 A 0 1 4 2 4

4. Question 5 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

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(ii) Describe the effects on a person when the gene for producing lactase is switched off. (lines 3 to 7)

(2)

(d) Suggest how people in some parts of the world have evolved tolerance because of their diet. (lines 8 to 10)

(2)

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(e) Explain how the symptoms of lactose intolerance are caused. (lines 11 to 15)

(4)

(Total for Question 5 = 14 marks)

4. Question 5 · part c.i · 3 marks · Working and answer

Working and answer

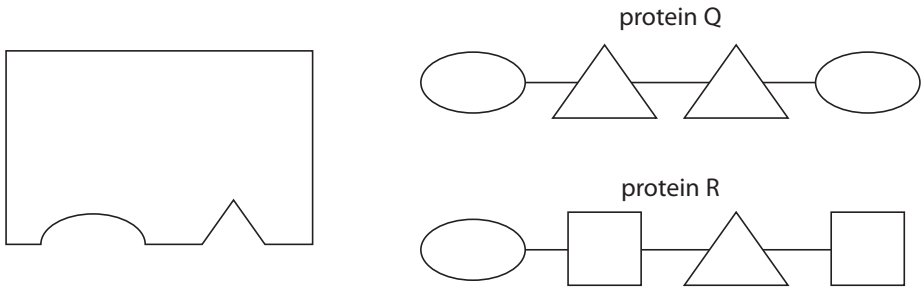
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5. Question 2 · part a.iii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

2 (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.



Enzyme P **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

(3)

amino acids	amylases	fatty acids	proteases
protein Q	protein R	starch	sugar

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

(1)

.....

(iii) Name the molecule that codes for the production of enzyme P.

(1)

.....

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5. Question 2 · part a.iii · 1 mark

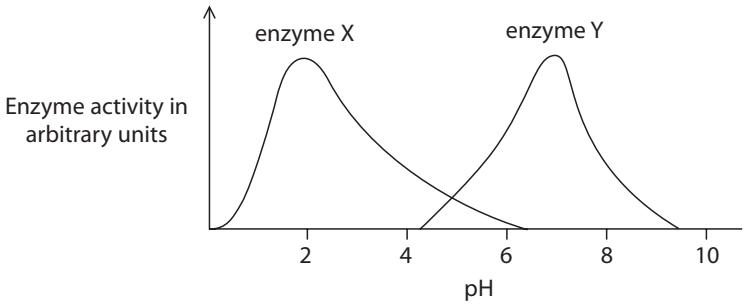
June 2023 Paper 2R

Further practice · Attempt part a.iii only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.



(i) Describe how the activity of enzyme X changes with pH.

(3)

(ii) Explain why no activity is shown for enzyme Y at pH2.

(2)

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P 7 2 5 8 8 A 0 6 2 4

5. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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Practice workbook · Apply the triplet code to bases, codons and amino acids

Converting DNA triplets to mRNA codons and tRNA anticodons, reading amino acids from a table, counting bases per amino acid, and counting possible codes as a power.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-APPLY-TRIPLET-CODE

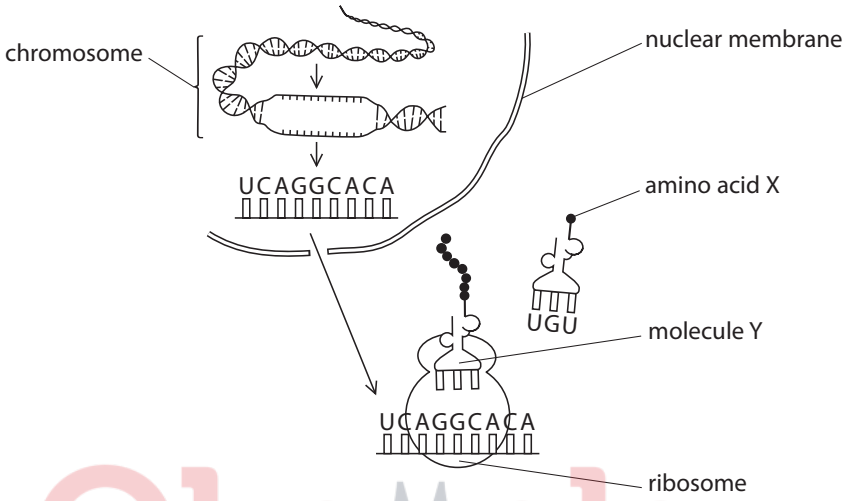
1. Question 6 · part c · 1 mark

June 2019 Paper 1R

Worked-example retry · Attempt part c only.

6 Protein synthesis has a number of stages.

The diagram shows some of the stages.



The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

Base triplet of DNA	Amino acid
AGT	serine
CCG	glycine
TGT	threonine
GTA	histidine
CAA	valine
TCC	argenine
ACA	cysteine

(a) Describe the function of the nuclear membrane.

(2)

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1. Question 6 · part c · 1 mark

June 2019 Paper 1R

Worked-example retry · Attempt part c only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

(1)

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

(1)

(d) Describe the process occurring on the ribosome.

(6)

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1. Question 6 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

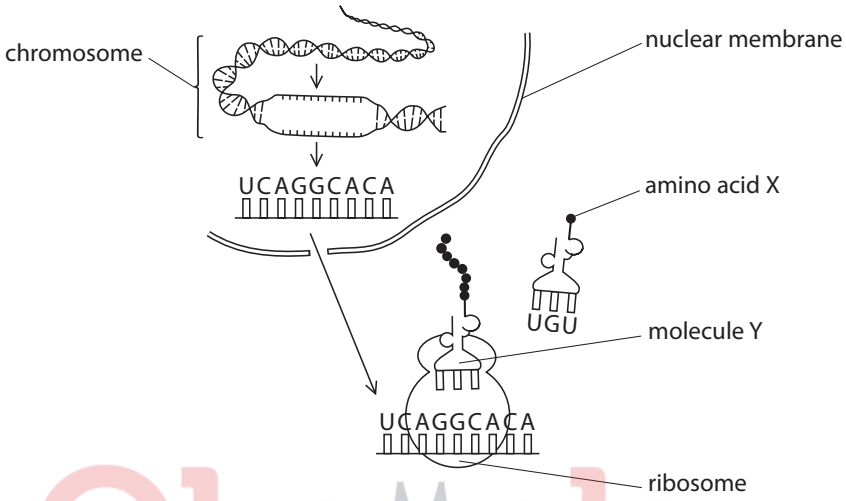
2. Question 6 · part b · 1 mark

June 2019 Paper 1R

Further practice · Attempt part b only.

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2. Question 6 · part b · 1 mark

June 2019 Paper 1R

Further practice · Attempt part b only.

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2. Question 6 · part b · 1 mark · Working and answer

Working and answer

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3. Question 8 · part a.iv · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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3. Question 8 · part a.iv · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

(v) Describe the structure of a nucleotide found in a DNA molecule.

(3)

(b) There are 20 different amino acids that can be joined to form a polypeptide chain.

The table gives some base sequences that code for five of these amino acids.

Code	Name of amino acid
AAT or AAC	asparagine
TAT or TAC	tyrosine
TTT or TTC	phenylalanine
CGT or CGG	arginine
CAT or CAC	histidine

Diagram 1 shows part of the gene that codes for a CFTR protein that functions normally.

Diagram 2 shows the same part of the gene with a mutation that causes cystic fibrosis.

AATATCATCTTTGGTGTTTCCTATGAT

Diagram 1

AATATCATCGGTGTTTCCTATGAT

Diagram 2

Describe the changes that occur in the gene and the protein that cause cystic fibrosis.

(3)

(Total for Question 8 = 14 marks)

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3. Question 8 · part a.iv · 2 marks · Working and answer

Working and answer

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4. Question 8 · part b · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

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This protein is responsible for transporting sodium ions across cell membranes using active transport.

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(2)

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4. Question 8 · part b · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

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(3)

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Diagram 1

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Diagram 2

Describe the changes that occur in the gene and the protein that cause cystic fibrosis.

(3)

(Total for Question 8 = 14 marks)

4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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5. Question 1 · part c · 2 marks

June 2023 Paper 2R

Further practice · Attempt part c only.

1 (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

(3)

Cell structure	Function		
	makes proteins	**contains DNA**	**allows substances in and out of cells**
nucleus			
ribosomes			
cell membrane			

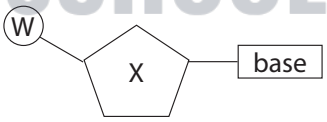
(ii) Give a structure found in bacterial cells.

(1)

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.



Name part W and part X.

(2)

W .....

X .....

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5. Question 1 · part c · 2 marks

June 2023 Paper 2R

Further practice · Attempt part c only.

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(ii) Describe how bases hold the two chains of nucleotides together.

(3)

.....

.....

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(c) The gene for the hormone insulin contains 153 bases.

Determine the number of amino acids in insulin.

(2)

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number of amino acids = ...

(Total for Question 1 = 11 marks)

5. Question 1 · part c · 2 marks · Working and answer

Working and answer

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6. Question 5 · part a · 2 marks

November 2023 Paper 1

Further practice · Attempt part a only.

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5 DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

(2)

number of codes = ...

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

(3)

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(ii) State **one** factor that can cause mutation.

(1)

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

(2)

6. Question 5 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · Use image size, actual size and magnification

A micrograph or diagram with a measurable length, a stated real size or magnification, or eyepiece and objective lenses.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-MAGNIFICATION

1. Question 3 · part b.i · 2 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b.i only.

3 (a) Describe what is meant by a specialised cell.

(2)

.....

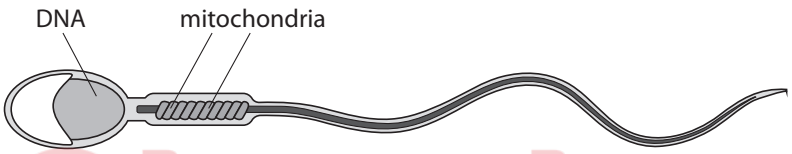
.....

.....

(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE



(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

(2)

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

(1)

.....

.....

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1. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

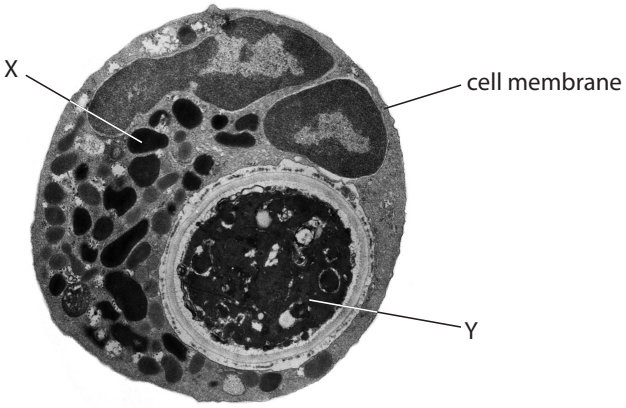
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2. Question 2 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

2 (a) The image is an electron micrograph of a human body cell.



(Source: © Science History Images/Alamy)

(i) The table gives information about parts of the human body cell.

Complete the table by giving the missing information.

(2)

Part	Name of part	Function
X		release energy from glucose
Y	nucleus	

(ii) Electron microscopes and light microscopes can both be used to view body cells.

Which of these is an advantage of using an electron microscope?

(1)

- ☒ **A** more cell structures can be seen with greater resolution
- ☒ **B** more cell structures can be seen with less resolution
- ☒ **C** fewer cell structures can be seen with greater resolution
- ☒ **D** fewer cell structures can be seen with less resolution

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2. Question 2 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

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(iii) The actual diameter of this human body cell before magnification is 0.05 mm.

Use information from the image to calculate the magnification of the cell.

(3)

magnification = ...

(b) Body cells produce carbon dioxide.

Describe how carbon dioxide passes from a body cell into the blood.

(2)

.....

.....

.....

.....

(Total for Question 2 = 8 marks)

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2. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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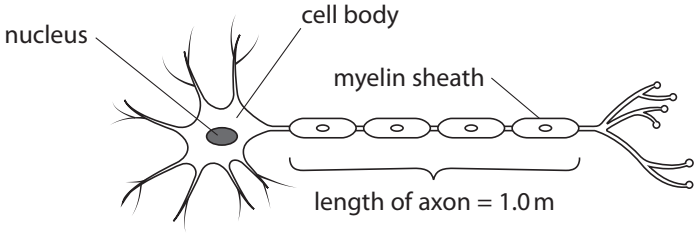
3. Question 5 · part a.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.i only.

5 The diagram shows a motor neurone.

not to scale



(a) (i) The actual diameter of the nucleus of the motor neurone is 12 µm.

A student observes the motor neurone using a ×50 microscope lens.

Calculate the diameter of the image seen after magnification.

Give your answer in millimetres.

(2)

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diameter = ... mm

(ii) Neurones transmit electrical impulses from one part of the body to another.

Suggest one function of the myelin sheath surrounding the axon.

(1)

.....

.....

(b) Nerve impulses travel at an average speed of 1.2×10^8 µm per second.

Hormones travel at an average speed of 1.5×10^6 µm per second.

(i) Explain this difference in speed.

(2)

.....

.....

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3. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 3 · part b.i · 1 mark

November 2020 Paper 1R

Further practice · Attempt part b.i only.

3 (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

(2)

	Steps
	select a suitable objective lens
5	adjust focusing wheel to obtain a clear image
1	remove cells from the inside of the cheek
	place microscope slide onto the microscope stage
	smear the cell sample onto a microscope slide

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

(1)

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

(1)

.....

.....

(iv) Give two ways that the student could avoid health hazards when working with living cells.

(2)

1 .....

.....

2 .....

.....

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4. Question 3 · part b.i · 1 mark

November 2020 Paper 1R

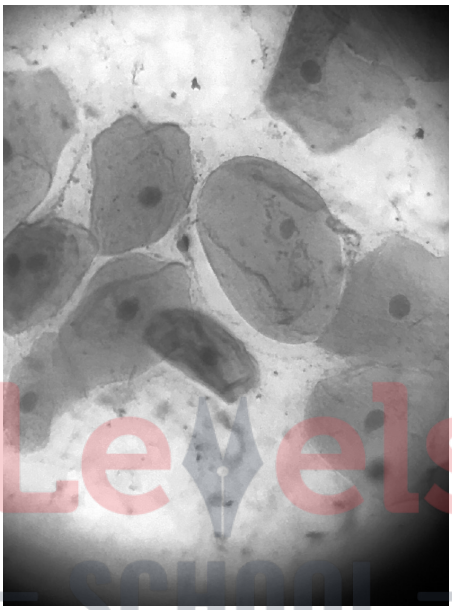
Further practice · Attempt part b.i only.

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(b) The photograph shows an image of a cheek cell seen under a light microscope using a 10× eyepiece lens and a 40× objective lens.



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© Esmiles/Shutterstock

(i) Calculate the total magnification used to view the cheek cell.

(1)

total magnification = ...

(ii) The actual diameter of the cheek cell shown in the photograph is 0.018 mm.

Convert the diameter of the cheek cell into micrometre (μm).
[1 μm = 10^{-6} m]

(2)

diameter = ... μm

(Total for Question 3 = 9 marks)

4. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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5. Question 5 · part c · 4 marks

June 2021 Paper 1

Further practice · Attempt part c only.

5 The diagram shows a sperm.



(a) (i) Describe the function of a sperm.

(2)

(ii) State two other structures, apart from mitochondria, that can be seen in the sperm if it is viewed using an electron microscope.

(2)

1 .....

2 .....

(b) Explain why there are a large number of mitochondria found in a sperm.

(3)

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5. Question 5 · part c · 4 marks

June 2021 Paper 1

Further practice · Attempt part c only.

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(c) The actual length of the sperm between X and Y on the diagram is 3 µm.

Calculate the magnification of the sperm shown in the diagram.

[1000 µm = 1 mm]

(4)

magnification = × .....

(Total for Question 5 = 11 marks)

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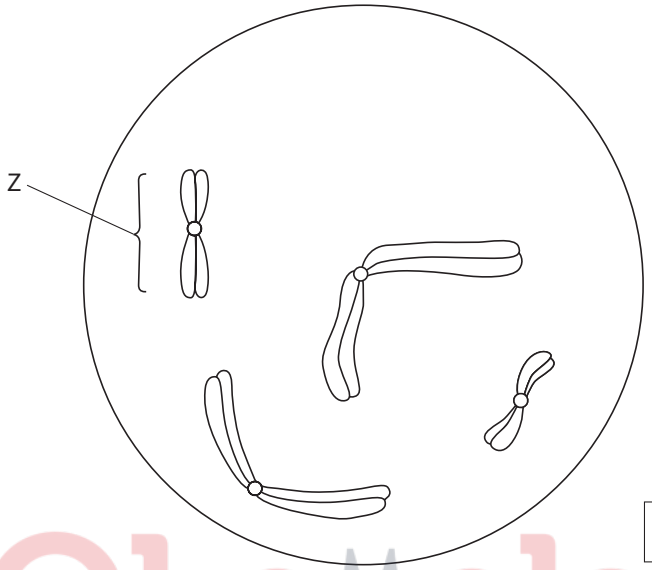
Working and answer

6. Question 7 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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P 6 7 0 6 0 R R A 0 1 6 2 0

6. Question 7 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

(b) (i) Which one of these molecules would be found in structure Z?

(1)

- ☒ **A** DNA
- ☒ **B** mRNA
- ☒ **C** tRNA
- ☒ **D** ribose sugar

(ii) Calculate the length of structure Z in micrometres.

[1 micrometre = 1×10^{-6} m]

(3)

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length of structure Z = ... micrometres

(c) Describe what is produced when a cell undergoes mitosis.

(2)

.....

.....

.....

.....

(Total for Question 7 = 11 marks)

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6. Question 7 · part b.ii · 3 marks · Working and answer

Working and answer

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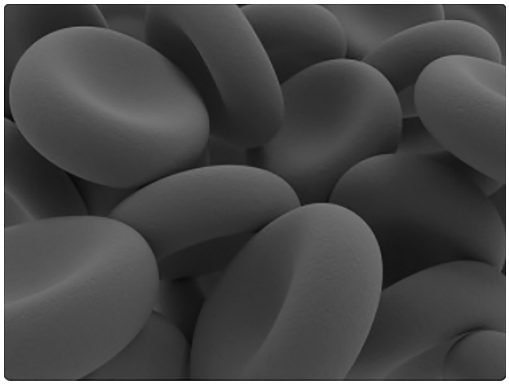
7. Question 1 · part b.iii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

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1 (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

(3)

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(ii) Explain how the structure of the red blood cell is adapted to its function.

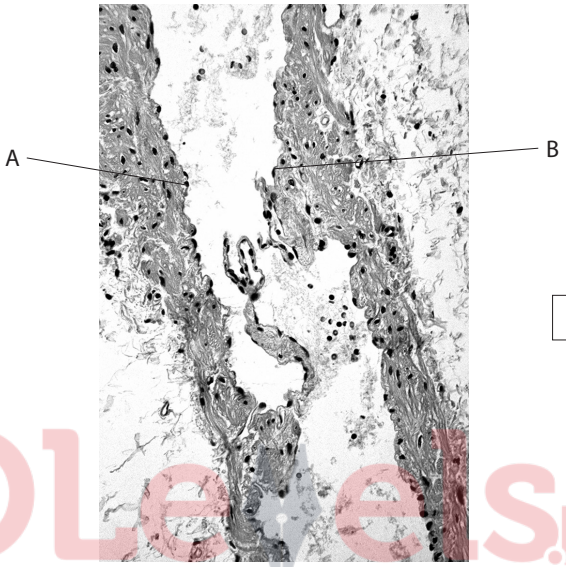
(4)

7. Question 1 · part b.iii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

(b) The photograph shows a section through a blood vessel in a leg.



magnification ×10

(Source: © DR. GLADDEN WILLIS, VISUALS UNLIMITED/SCIENCE PHOTO LIBRARY)

(i) State the type of blood vessel shown in the photograph.

(1)

(ii) Explain your choice of blood vessel.

(2)

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7. Question 1 · part b.iii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

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(iii) Calculate the actual distance between points A and B.

(2)

distance = ... mm

(Total for Question 1 = 12 marks)

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Working and answer

8. Question 1 · part a.ii · 3 marks

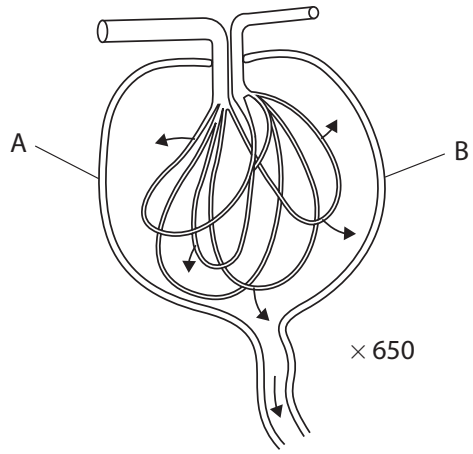
January 2022 Paper 2

Further practice · Attempt part a.ii only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

8. Question 1 · part a.ii · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

Time after drinking in hours	Volume of urine in cm^3	
	day 1 (20 °C)	day 2 (30 °C)
0.5	70	50
1.5	175	70
2.0	300	200
2.5	250	190
3.0	210	150
4.0	100	50

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8. Question 1 · part a.ii · 3 marks · Working and answer

Working and answer

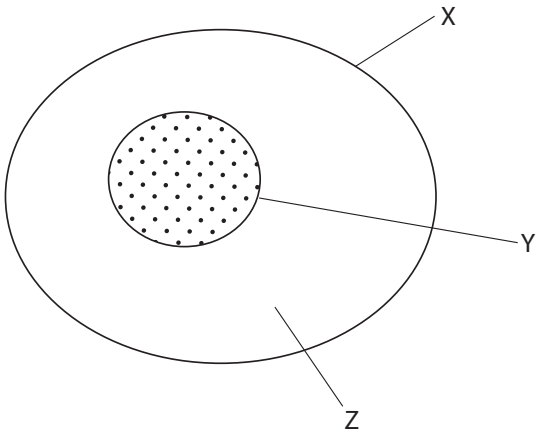
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9. Question 1 · part c.ii · 3 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

1 (a) The diagram shows a human body cell.



(i) Name the parts X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

(1)

.....

(iii) Explain how body cells are organised to form organs.

(2)

.....

.....

.....

.....

.....

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9. Question 1 · part c.ii · 3 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

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(b) The boxes list three processes that transport substances in and out of cells.

Complete the diagram by drawing one straight line from each process to its correct definition.

(3)

Process	Definition
	requires energy to move molecules against a concentration gradient
osmosis	
	water moves from a high water potential to a low water potential
diffusion	
	water moves from a low water potential to a high water potential
active transport	
	molecules move down a concentration gradient

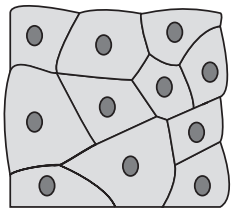
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9. Question 1 · part c.ii · 3 marks

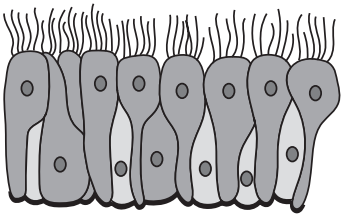
June 2022 Paper 1R

Further practice · Attempt part c.ii only.

(c) (i) The diagram shows two types of epithelial tissue, A and B.



Tissue A



Tissue B

(Source: © Timonina/Shutterstock)

Give the name of each type of epithelial tissue.

(2)

A .....

B .....

(ii) An epithelial cell has an actual width of 60 micrometres (μm).

The cell is magnified 40 times.

Calculate the image width of this cell in mm.

[1 mm = 1000 μm]

(3)

image width = ... mm

(Total for Question 1 = 14 marks)

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9. Question 1 · part c.ii · 3 marks · Working and answer

Working and answer

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10. Question 4 · part a.ii · 2 marks

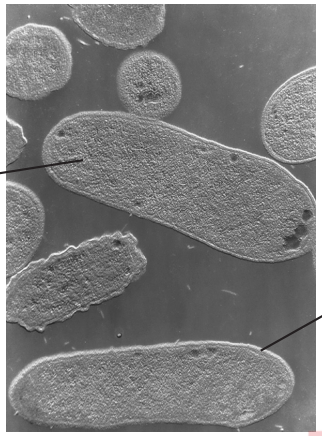
January 2023 Paper 1

Further practice · Attempt part a.ii only.

4 Typhoid is caused by *Salmonella typhi* bacteria.

These bacteria are spread by houseflies.

(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.



(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

(1)

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

(2)

actual length = ... µm

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10. Question 4 · part a.ii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

(i) Flies in towns and cities carry more bacteria than flies that are found in the countryside.

Suggest a reason for this observation.

(1)

(ii) One species of bacterium identified is *Helicobacter pylori* which causes stomach ulcers.

Explain how houseflies could increase the transmission of *Helicobacter pylori*.

(3)

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10. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

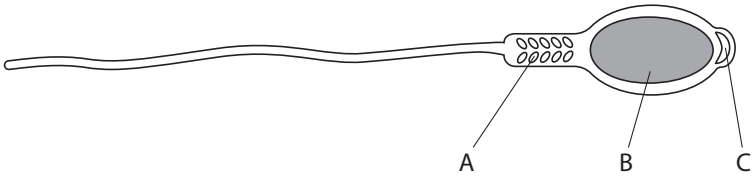
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11. Question 6 · part a.iii · 4 marks

November 2023 Paper 2

Further practice · Attempt part a.iii only.

6 The diagram shows a human sperm.



(a) (i) Name the parts A, B and C.

(3)

A .....

B .....

C .....

(ii) Describe the function of part C.

(3)

.....

.....

.....

.....

.....

.....

(iii) The actual length of the sperm is 50.0 micrometres (μm).

Calculate the magnification of the diagram.

[1 mm = 1000 μm]

(4)

magnification = ...

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P 7 4 4 9 4 A 0 1 4 2 0

11. Question 6 · part a.iii · 4 marks

November 2023 Paper 2

Further practice · Attempt part a.iii only.

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(b) Describe the function of the structure labelled A in the process of fertilisation.

(3)

(Total for Question 6 = 13 marks)

Working and answer

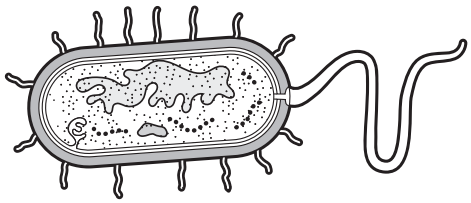
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12. Question 1 · part b.ii · 3 marks

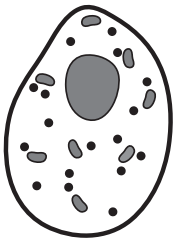
June 2024 Paper 1R

Further practice · Attempt part b.ii only.

1 (a) The diagrams show a bacterial cell and a human skin cell.



Bacterial cell



Human skin cell

(i) Give **two** similarities between the bacterial cell and the human skin cell.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** differences between the bacterial cell and the human skin cell.

(2)

1 .....

.....

2 .....

.....

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12. Question 1 · part b.ii · 3 marks

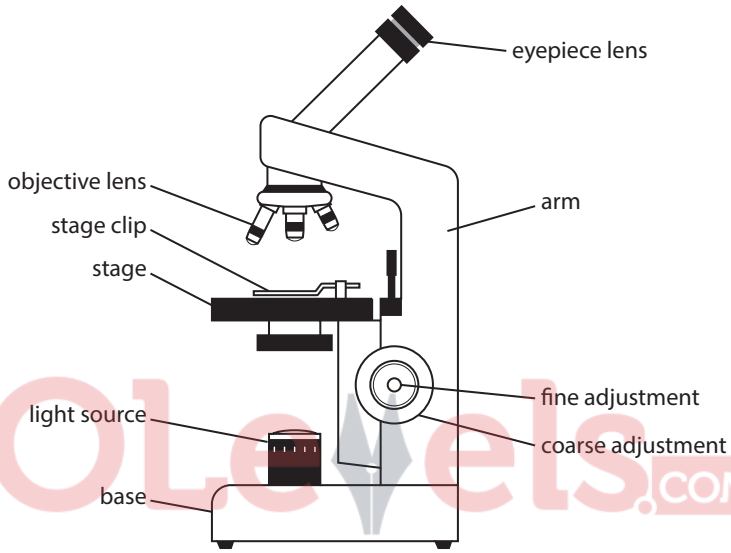
June 2024 Paper 1R

Further practice · Attempt part b.ii only.

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(b) Many structures in cells can be viewed more clearly using a light microscope.

The diagram shows a light microscope.



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(i) Draw **one** straight line from each part of the light microscope to its function.

(3)

Part of microscope	Function
objective lens	holds the microscope slide in place
stage clip	moves the stage to produce a clear, focused image
fine adjustment	magnifies the sample
	keeps the microscope steady

12. Question 1 · part b.ii · 3 marks

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

(ii) The image size of a bacterial cell viewed using an electron microscope is 5.0 cm.

The bacterial cell has been magnified 50 000 times.

Calculate the actual size of this bacterial cell in mm.

image size = actual size × magnification

(3)

actual size of bacterial cell = ... mm

(Total for Question 1 = 10 marks)

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Working and answer

13. Question 4 · part b.ii · 3 marks

June 2025 Paper 1

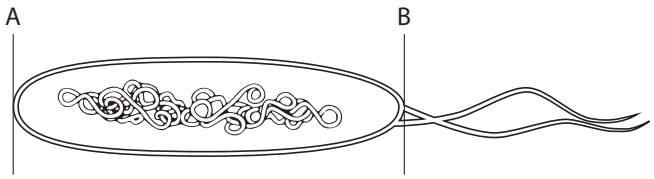
Further practice · Attempt part b.ii only.

4 (a) State what is meant by the term **pathogen**.

(1)

.....

(b) The diagram shows a pathogenic bacterium.



(i) Label **three** structures shown on the diagram.

(3)

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

(3)

magnification = × .....

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13. Question 4 · part b.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part b.ii only.

(c) The effectiveness of a vaccine in preventing disease was studied over six months.

Two different groups of people were used. One group of people were aged between 45 and 64 years and the second group were aged over 65 years.

The results are shown in the table.

	Percentage effectiveness against disease (%)	
Time since vaccination in months	**45 to 64 years**	**over 65 years**
1	90	80
2	85	78
3	80	75
4	68	58
5	60	50
6	50	40

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13. Question 4 · part b.ii · 3 marks · Working and answer

Working and answer

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